

Concord is Insatiable Agree¹

Heterogeneous Case and Phi Patterns in Slavic and Finnic

1 Introduction

Minimalism strives towards explaining the richness of syntactic phenomena with a minimal number of theoretical constructs. Two operations are widely assumed to be irreducible, although their exact nature and possible variants are still a matter of lively debate: Merge (for building structures) and Agree (for establishing dependencies between constituents). Also the need for some kind of locality theory, such as Phase Theory, is widely accepted. Other mechanisms are treated as suspect, and the goal of reducing them to Merge, Agree, and/or Phases is generally deemed worth pursuing.

One area that has resisted such reduction is concord, understood as agreement between nominal constituents, typically in case, number and gender. Despite considerable similarities to agreement in ϕ features between nominal and verbal constituents (e.g., Baker 2008), universally analyzed via Agree, it is commonly assumed that other, more specialized feature percolation mechanisms are needed in order to successfully account for nominal concord (e.g., Norris 2014, 2018, Bayırılı 2017).

The first aim of this paper is to show that recent interaction/satisfaction approaches to Agree (Preminger 2014, Deal 2015, 2023, 2024, Clem 2019, 2023, Clem and Deal 2024; briefly introduced in §2 below) predict nominal concord almost out of the box; §3. The second aim is to demonstrate that – given the ϕ -feature-stacking aspect of Agree (e.g., Deal 2015, Hammerly 2020, Clem 2023, Clem and Deal 2024, Grishin 2024) and the well-known phenomenon of case stacking (e.g., Richards 2013, Pesetsky 2013, Levin 2017, Chen 2018 and references therein) – this account naturally extends to the puzzling heterogeneous vs. homogeneous case patterns in Slavic numeral–noun constructions (e.g., Babby 1985, 1987, Franks 1994, 1995); §4. In particular, this approach makes it possible to explain the baffling pattern of ϕ -features in Russian paucal numeral constructions (e.g., Mel’čuk 1985, Pesetsky 2013); §5. The resulting analysis is extended to the Finnic – especially, Estonian – data which motivated a prominent specialized approach to concord (Norris 2014, 2018); §6. The conclusion is that concord is simply insatiable Agree and does not necessitate any special theoretical mechanisms; §7.

2 Agree

Some recent approaches to Agree reject the idea that the aim of Agree is to value uninterpretable features on the penalty of derivation failure (Chomsky 2000, 2001). On these approaches, Agree failure does not lead to ungrammaticality, but results in “default agreement” (Preminger 2014). Moreover, given the richness and diversity of agreement phenomena across languages, Agree is taken to be a relatively robust operation, specifiable both in terms of interaction (which features of a goal the probe interacts with) and satisfaction (which features of a goal, if any, make the probe stop looking for further goals) (Deal 2015 *et seq.*). Importantly, probing multiple goals may result in a number of feature bundles on the probe (as well as on the goal), with the effect that the collection of features on a constituent has a more articulate structure than just an unordered set of features (e.g., Deal 2015, Hammerly 2020, Clem 2023, Grishin 2024).

These aspects of the relevant approaches to Agree may be illustrated with Clem and Deal’s (2024) analysis of ergative in Shawi (spoken in Peru). In this language, ergative only appears when the subject is at least as high as the object on the person hierarchy (1 > 2 > 3, where ‘>’ means ‘higher than’), and may then be accompanied by explicit realization of the object’s features on the subject. This is shown in (1)–(2), originally from Bourdeau 2015 (Clem and Deal 2024: 6, (6a)–(7a)):²

- | | |
|--|---|
| (1) I’wara ka-ri-nke pera-ra(-w)-nke. (CBT)
yesterday 1-ERG-2.O call-IND-1-2.O
‘I called you yesterday.’ | (2) I’wara kema(*-ri) nu’wi-r-an-ku. (CBT)
yesterday 2(*-ERG) tell.off-IND-2-1.O
‘You told me off yesterday.’ |
|--|---|

In (1), a 1st person subject acts on a 2nd person (i.e., lower) object. As a result, the subject is marked with ERG (-ri) and, optionally, with ϕ features of the object (-nke). In (2), a 2nd person subject acts on a 1st person (i.e., higher) object, and the subject cannot be marked with either.

¹This paper was written when I was a Visiting Professor at MIT (September 2024 – April 2025) and enjoyed the stimulating environment of linguistic departments of MIT and Harvard. It benefited immensely from many discussions with David Pesetsky, as well as from interactions with Peter Grishin, Jonathan Bobaljik, Norvin Richards, Tanya Bondarenko, Elise Newman, Sabine Iatridou, and graduate students both at MIT and at Harvard. Parts of this paper were presented at various MIT seminars: LingLunch, Syntax Square, and MorPhun. Many thanks for comments on earlier versions of this paper are also due to Agnieszka Patejuk and Sebastian Zawada of the Formal Linguistics Group at ICS Polish Academy of Sciences. My stay at MIT was possible thanks to the Bekker NAWA numer BPN/BEK/2023/1/00064.

The current status of this paper is an unpublished draft. I hope to add the conclusion and technical appendices in early 2026 and to submit it – or extend to a book? – soon after that. Hence, any comments at this stage will be very welcome.

²All examples are annotated with 3-letter ISO 639-3 codes of respective languages: AMI = Amis, CBT = Shawi, DAN = Danish, DEU = German, EST = Estonian, FIN = Finnish, GOT = Gothic, GYD = Kayardild, KHA = Khasi, LBZ = Lardil, MIQ = Misikitu, POL = Polish, RUS = Russian, VMA = Martuthunira. Glossing conventions and abbreviations generally follow the Leipzig Glossing Rules, with o in (1)–(2) additionally indicating agreement with object. Additional abbreviations (e.g., of Estonian cases in §6.1) may be found in https://en.wikipedia.org/wiki/List_of_glossing_abbreviations#Grammatical_abbreviations.

Clem and Deal (2024) – assuming cyclic Agree (Béjar and Řezáč 2009) and the interaction/satisfaction framework (Deal 2015, 2024) – analyze (1)–(2) by postulating that a probe on v , which is situated between the subject and the object, is specified as $[\text{INT}:\phi, \text{SAT}:\text{SPEAKER}]$: it copies all ϕ features from any element it interacts with, but this interaction stops upon encountering SPEAKER , i.e., 1st person. This is what happens in (2); the probe flags the object with its features (only v , at this stage) and simultaneously copies over (agrees with) the object’s ϕ features. As the object is 1st person, the $\text{SAT}:\text{SPEAKER}$ condition on the probe is met and probing stops here. By contrast, in (1) the object is not 1st person, so probing does not stop here; a projection of the same v probe additionally flags the subject with its features – which now crucially include the object’s ϕ features in addition to the categorial v feature – and copies the subject’s ϕ features.

The resulting features of the key items in (1)–(2) are schematically shown in (3)–(4), where ϕ_o refers to the object’s ϕ features, and ϕ_s to the subject’s features.³

- | | |
|--|--|
| <p>(3) transitive 1→2 in Shawi (as in (1)):</p> <p>S (ERG): $[\text{D}, \phi_s]$ $[v, \phi_o]$</p> <p>v: $[v] [\phi_o] [\phi_s]$</p> <p>O (ABS): $[\text{D}, \phi_o] [v]$</p> | <p>(4) transitive 2→1 in Shawi (as in (2)):</p> <p>S (ABS): $[\text{D}, \phi_s]$</p> <p>v: $[v] [\phi_o]$</p> <p>O (ABS): $[\text{D}, \phi_o] [v]$</p> |
|--|--|

The first bundle on each item represents its native features, while subsequent bundles are the result of consecutive probe–goal interactions. The bundle that is crucial in this analysis is the boxed $[v, \phi_o]$ on the subject in (3), containing ϕ_o – the ϕ features of the object. It is the presence of two sets of ϕ features – ϕ_s and ϕ_o – that is expounded in the context of D as the ergative suffix *-ri*.

Clem and Deal (2024: §5) extend this general approach to accusative languages. There, the relevant probe is assumed to be above the subject, on T, not on v situated between the subject and the object, as in ergative languages. Hence, the probe on T first interacts with the subject and then flags the object with the subject’s ϕ features; see (5). On this view, ACC is keyed to the presence of two ϕ feature bundles – ϕ_o and ϕ_s – on a DP object.

- (5) transitive in accusative languages:
- T: $[\text{T}] [\phi_s] [\phi_o]$
- S (NOM): $[\text{D}, \phi_s] [\text{T}]$
- O (ACC): $[\text{D}, \phi_o]$ $[\text{T}, \phi_s]$

While the probe that Clem and Deal (2024) postulate in their analysis of Shawi is $[\text{INT}:\phi, \text{SAT}:\text{SPEAKER}]$, they note that the more common languages, in which the case of one argument does not depend on the person of another, require a probe with no satisfaction constraints: $[\text{INT}:\phi, \text{SAT}:-]$. Such a probe is insatiable: it interacts with all accessible ϕ -bearing elements (e.g., Clem 2023, Deal 2024). In the next section, we will see that such insatiable probes may be responsible for concord.

3 Concord

3.1 Extended Nominal Projections

It turns out that the view of Agree presented in the previous section predicts the possibility of concord on almost any approach to the structure of nominal phrases.⁴ The only crucial assumption that needs to be made is the prevalent assumption that maximal nominal projections are headed by a functional head such as D or K (on which more below), i.e., that each maximal nominal projection has the structure in (6), where F is a nominal functional head capable of hosting the case feature.

Maximal extended nominal projection:

- | | |
|--|--|
| <p>(6) minimal assumption, schematically:</p> <div style="text-align: center; margin-top: 10px;"> </div> | <p>(7) assumed here (to be extended):</p> <div style="text-align: center; margin-top: 10px;"> </div> |
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³In (3)–(5), nominal constituents are assumed to be of category D, while S and O indicate subject and object.

⁴In this paper, *nominal phrase* refers to any (extended) nominal projection, including DP and KP, not just an NP.

The dominant view since Abney 1987 is that F is D(eterminer). However, it is also frequently noted that another functional level is needed, corresponding to morphological case, i.e., that F is K(ase) (e.g., Lamontagne and Travis 1987, Bittner 1994, Bittner and Hale 1996b,a, Levin 2015; cf. Caha 2009 *et seq.*). One argument for this assumption is that some languages seem to overtly realize both D and K, in accordance with their general headedness pattern, as in (8) (Khasi – a head-initial Austroasiatic language spoken in India) and (9) (Misikitu – a head-final Misumalpan language spoken in Nicaragua), both from Bittner 1994: 4.

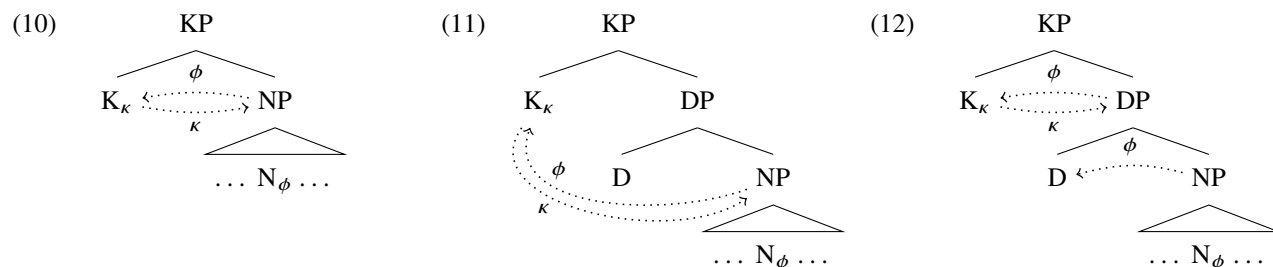
- (8) Ka la yo”ii [ya ’u khlaa]. (KHA)
 she PST see ACC the tiger
 ‘She saw the tiger.’
- (9) [Waitna ba] [sula ba ra] kaik-an. (MIQ)
 man the deer the ACC see-PST.3SG
 ‘The man saw the deer.’

The minimal structure of nominal projections in such examples is (7), with all heads overtly realized. Hence – even if, for the purposes of this paper, F in (6) could be any functional head capable of hosting case features – we will follow the increasingly common view that maximal nominal projections are indeed KPs, as in (7).⁵ However, we will not prejudice the NP/DP issue, i.e., we will remain agnostic about whether the DP projection is present in all (or any) languages.

3.2 Agree in Nominal Projections

What are the *minimal* assumptions that need to be made about Agree in maximal nominal projections headed by a case-bearing functional head, e.g., K? Given that 1) such maximal KP projections bear ϕ features (as is clear from standard binding and agreement facts), that 2) the source of ϕ features is N or projections just above it rather than K (a standard assumption), and that 3) there are no specialized feature percolation mechanisms (a methodological assumption we make here), K must Agree with its nominal complement in order to receive ϕ features. Then, since the K head is featurally identical with its projections,⁶ these ϕ features are also present on KP. Hence, K must be assumed to host a probe interacting with a c-commanded constituent bearing ϕ features; such a probe may be minimally specified as: [INT: ϕ , SAT: ϕ].

Assuming, as in the work on Agree summarized in §2 above, that probes flag their goals with their own (the probes’) features, another result of this Agree is the transfer of K’s case feature(s), κ , to its complement. This minimum exchange of features is schematically illustrated in (10), which assumes no DP layer.⁷



If a DP layer is present, there are two possibilities, as schematically shown in (11)–(12). The first is that D does not Agree with its NP complement, so the closest constituent bearing ϕ features that is c-commanded by K is the NP, with which K exchanges features directly (see (11)). In this case, the NP (and, hence, also its N head) ends up with the case features κ , but the DP (and, hence, the D) remains case-less. The second possibility is that D does Agree with NP, in which case the closest constituent bearing ϕ features that is c-commanded by K is the DP (see (12)). In this case, the DP (and, hence, the D), but not the NP (or the N), receives κ .

Note that none of these possibilities allows for concord; in all, exactly two categories bear κ : K(P) and either N(P) alone (in (10)–(11)) or D(P) alone (in (12)). Assuming that, if a feature of a head H is morphologically realized at all, it may be realized either on H itself or – as a phrasal affix – on H’s maximal projection (but normally not both), up to two exponents of case in such structures are predicted, but also possibly one (for example, if K is phonologically empty and κ is not realized phrasally on KP) or even none (if category-specific case impoverishment rules are in play). An example of

⁵Norris 2022: 6 notes that “KP has become so widely assumed that it has been used in published, peer-reviewed research without citation for over a decade...”.

⁶This follows from the common Minimalist assumption of Bare Phrase Structure (Chomsky 1995); cf. analogous assumptions built into LFG (Kaplan and Bresnan 1982, Bresnan *et al.* 2016, Dalrymple *et al.* 2019, Dalrymple 2023) and – with some caveats – HPSG (Pollard and Sag 1987, 1994, Müller *et al.* 2024).

⁷Intrinsic features of particular heads are marked as subscripts.

a language with nominal structures as in (12) may be English, where nominal agreement and case are realized only on (some) Ds (cf. *this book* vs. *these books* and *he* vs. *him*).

While there are many languages with just one or two possible exponents of case within any nominal projection, it is often not easy to decide which of the structures postulated above are realized in a given language. The matter seems relatively clear in languages such as Diyari (Australian), where case is expounded on a determiner and on the final word of the nominal projection, whether it is a noun or an adjective, e.g., *nhu-lu karna pirna-li* ‘he-ERG man big-ERG’, i.e., ‘the big man’ in ergative (Austin 1981 apud Blake 2001: 101, (21)). Such languages probably have nominal structures as in (12), with κ expounded on D and – as a phrasal suffix – on KP. However, even here another possible analysis presents itself: the structure might be as in (11), with D head-moving to K and realizing the determiner–case bundle, KP realizing κ as a phrasal affix, and with nouns being morphologically impoverished for case. We leave such matters for language-specific research, but – for concreteness – we will assume that D normally Agrees with NP, as in (12), given that D expounds the NP’s gender and number in many familiar languages such as French or German.

We now move to the consideration of concord languages, in which case and ϕ features have a theoretically unbounded number of exponents within a single nominal projection.

3.3 Concord is Insatiable Agree

In concord languages, case and ϕ features are typically expounded on the noun and all agreeing modifiers, regardless of their number. This is illustrated with the following attested examples from Polish, each with five inflectionally adjectival modifiers⁸ agreeing with the noun in case (nominative in (13) and genitive in (14)) and in ϕ features.⁹

- (13) [Jakiś drugi taki sam mały chłopiec] podskoczył. . .
some.M.SG.NOM second.M.SG.NOM such.M.SG.NOM same.M.SG.NOM little.M.SG.NOM boy.M.SG.NOM] jumped.M.3SG
‘[Some little boy just like the other one] jumped. . .’ (POL)
- (14) . . . spełnienie [tej swojej cichej wielkiej skrywanej namiętności]. . .
fulfillment.N.SG.NOM this.F.SG.GEN self.F.SG.GEN quiet.F.SG.GEN great.F.SG.GEN hidden.F.SG.GEN passion.F.SG.GEN
‘. . . fulfillment [of that quiet great hidden passion of her/his]. . .’ (POL)

In the previous subsection, the probe on K was assumed to be specified as [INT: ϕ , SAT: ϕ]. However, another possibility predicted by recent approaches to Agree is that the probe is insatiable: [INT: ϕ , SAT:–]. We claim that it is this insatiable Agree that directly accounts for concord in languages such as Latin, Polish and Estonian.

To see this, recall Clem and Deal’s (2024) analysis of Shawi ergative (§2). There, a v probe first flags a constituent with its categorial v feature and “agrees” with that constituent in ϕ features, i.e., copies those features over. Subsequently, the same probe flags another constituent, but now not just with v , but with *all* its features, i.e., with the whole [v , ϕ] bundle.

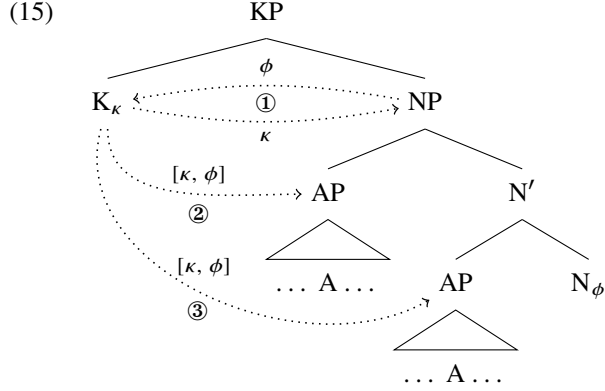
We propose that exactly the same mechanism is responsible for concord, the only difference being that K flags its nominal complement with its case, κ , rather than only with its category K. In the process, it copies over the ϕ features of the nominal complement, and any subsequent probing amounts to flagging with the [K, κ , ϕ] bundle.¹⁰ This way not only the particular case κ , but also the noun’s ϕ features end up on all accessible constituents within the maximal nominal projection. This is schematically illustrated in (15), with two adjectival modifiers.¹¹ Once this tree is fully built, all constituents contain both ϕ and κ among their features.

⁸By “inflectionally adjectival modifiers” we understand not only standard APs, but also, for example, (phrases headed by) adjectival participles, as well demonstrative pronouns, which may or may not be reasonably analyzed as D(eterminer)s, but which, like prototypical adjectives, inflect for case, gender and number in languages such as Polish.

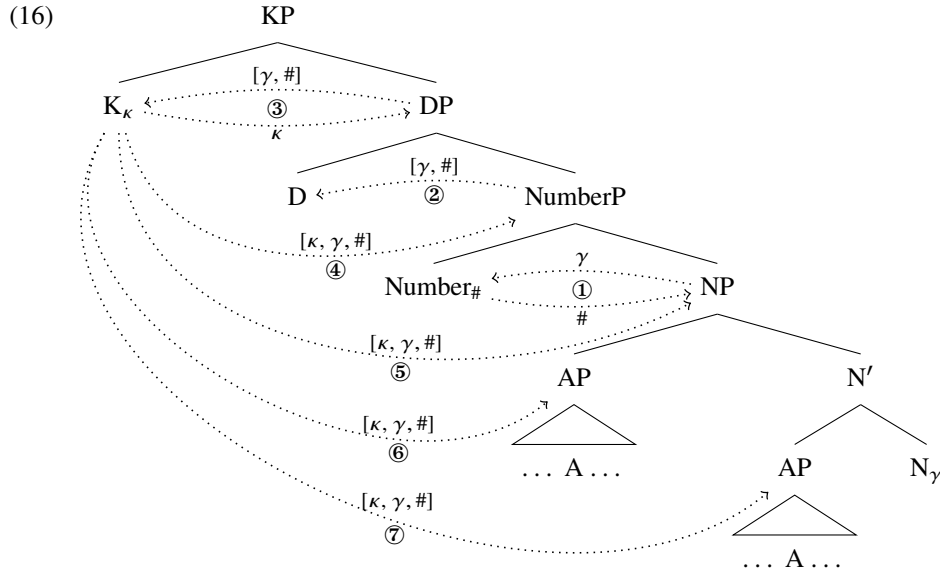
⁹All Polish examples are from the National Corpus of Polish (NCP; Przepiórkowski *et al.* 2012; <https://nkjp.pl/>); some are slightly simplified in ways that do not affect corresponding claims. Note that the development of NCP ended in 2011, which guarantees that it does not contain AI-generated texts.

¹⁰A somewhat similar mechanism is assumed – but not fully formalized – in Pesetsky 2013: 32. Since category flagging does not matter for most of this paper, we will usually omit categories in such feature bundles. See Philippova 2018 for a phenomenon that may be analyzed as category flagging expounded independently of case flagging.

¹¹We make here a simplifying assumption that Agree only targets maximal projections, e.g., NP but not N’ or N in (15). (Technically, this can be ensured by requiring that any given probe does not target the same goal twice and that constituents with the same – shared – labels count as the same goals.) However, nothing seems to hinge on this assumption. In its absence, Agree would also target non-maximal projections, attempting to flag them with a feature bundle that they already have (via identity with the higher projection that has already been flagged). If feature bundles do not constitute sets but rather stacks, as assumed below, this might lead to the presence of multiple identical feature bundles on a label. However, such bundles would never be multiply expounded, given the widespread haplology of affixes of the same type (see, e.g., Richards 2013: 50–51), so their presence on the stack does not seem to be a problem. Nevertheless, if this multiple presence of identical bundles turns out to be technically problematic in some way, a syntactic “feature bundle haplology” rule may be assumed, which prohibits adding to the stack a bundle that is already present on top of the stack.



In (15), the only functional head is K, but concord works similarly in more articulated structures. This is illustrated in (16), which includes a DP layer and assumes that grammatical number, #, is introduced by a separate functional head (Ritter 1991), while the lexical N is intrinsically specified for grammatical gender, γ . (This structure could be further extended by splitting lexical N into root and nominalizer, by including a separate functional head introducing grammatical gender, etc.)



Here, in the first interesting step, Number Agrees with the NP, resulting in bidirectional passing of relevant ϕ features: γ from NP to Number and # from Number to NP. At this point Number contains both gender and number, and – hence – so does its maximal projection. Then, D Agrees with NumberP, copying over its ϕ features γ and #. In the third relevant step, K Agrees with the DP, resulting in the strengthening of its κ flag to $[\kappa, \gamma, \#]$. In further steps, K insatiably Agrees with other constituents it c-commands. As a result, all constituents bear the $[\kappa, \gamma, \#]$ bundle, resulting in concord.¹²

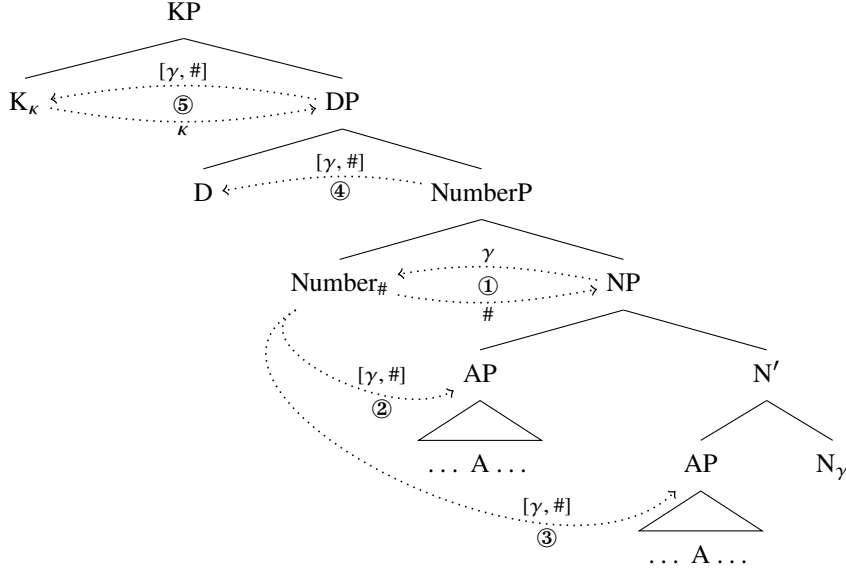
3.4 Typology of Concord

Given that the KP projection is universally taken to be the maximal nominal projection, certainly above the Number projection (see again (16) above), the proposed approach to concord makes a prediction: in languages with both overt case and overt ϕ features, concord may target either both or ϕ features alone, but not case alone. This is why. In the previous subsection, §3.3, we saw how an insatiable probe on K results in full concord in case and ϕ features. If there is no insatiable probe, there is no concord, as discussed in §3.2. However, there is also another possibility, namely, that the probe on K is immediately satiable, but the one on D – or Number, if present – is not. Then, only ϕ concord results.

Let us illustrate this possibility with (17), i.e., a structure just like (16), but with an insatiable probe present on Number instead of K. Essentially the same reasoning applies if the insatiable probe is present on D, and if only one of the DP and NumberP layers is present.

¹²Technically, Number(P) and N(P) also bear the earlier bundle $[\gamma, \#]$, unless some kind of “feature bundle haplology” rule (see fn. 11) is assumed. This is not a problem, given the common mechanisms of expounding the outermost (most recent) or the most marked feature bundle, discussed in §4.

(17)



As above, in the first interesting step, Number Agrees with the NP, resulting in bidirectional passing of relevant ϕ features: γ from NP to Number and $\#$ from Number to NP. However, since the Number probe is insatiable here, it flags both APs with γ and $\#$ in subsequent two steps, resulting in ϕ concord. The final two steps are identical to the corresponding two steps in (16), but – unlike there – they are not followed here with K flagging all maximal projections with $[\kappa, \gamma, \#]$, as the K probe is now immediately satisfied when it Agrees with the DP. This way, κ is only present on D(P) – and may be expounded there – but not on APs or directly on the NP.

What the current system does not make possible to derive is the opposite situation, i.e., case concord with no ϕ concord. This is because, if there is case concord, the probe on K must be insatiable, but if it is, it should flag all constituents not only with case, but also with the ϕ features it copied from its nominal sister in the first probing step.

Typological considerations confirm the correctness of this prediction. As discussed in Bayırılı 2017: ch. 2, there are many languages with complete case and ϕ concord and multiple languages with pure ϕ concord despite the presence of a case system, but no known languages with pure case concord despite the presence of overt ϕ exponents.¹³

Note that no additional assumptions were needed to derive this typological generalization, apart from the independent and universally adopted assumption that KP is the maximal nominal projection, higher than any projections introducing ϕ features.^{14,15} The next subsection provides another striking confirmation of the approach to concord proposed here, based on overt case stacking in Lardil.

3.5 Phases and Case Stacking

The analysis above seems to predict that absolutely all components of a nominal phrase must agree in case and ϕ features, while in fact such phrases may contain other nominal projections, bearing their own case and ϕ features in many languages, as in the Polish example (18), in which one of the constituents of a feminine dative nominal phrase is masculine genitive.

- (18) Nie poddawajcie się [tej zgubnej porze roku]. (POL)
 NEG surrender.IMP.2PL this.F.SG.DAT fatal.F.SG.DAT season.F.SG.DAT year.M.SG.GEN
 ‘Do not surrender to [this fatal time of year.M.GEN].F.DAT.’

We adopt the very common assumption that maximal nominal projections – here, KPs – are *phases*.¹⁶ We also assume the most common *strong* version of the Phase Impenetrability Principle (PIC), that of Chomsky 2000, 2007, 2008, upon which the sister of the phase head, here K, is spelled out as soon as that phase head fully projects, here to KP. However, all reasoning in this section is also valid if the *weak* PIC of Chomsky 2001 is adopted, upon which the sister of a phase

¹³Note that it is theoretically possible that a language has grammatical case but no grammatical number or grammatical gender (just as, e.g., Turkish has no grammatical gender), in which case pure case concord is a possibility predicted by our proposal.

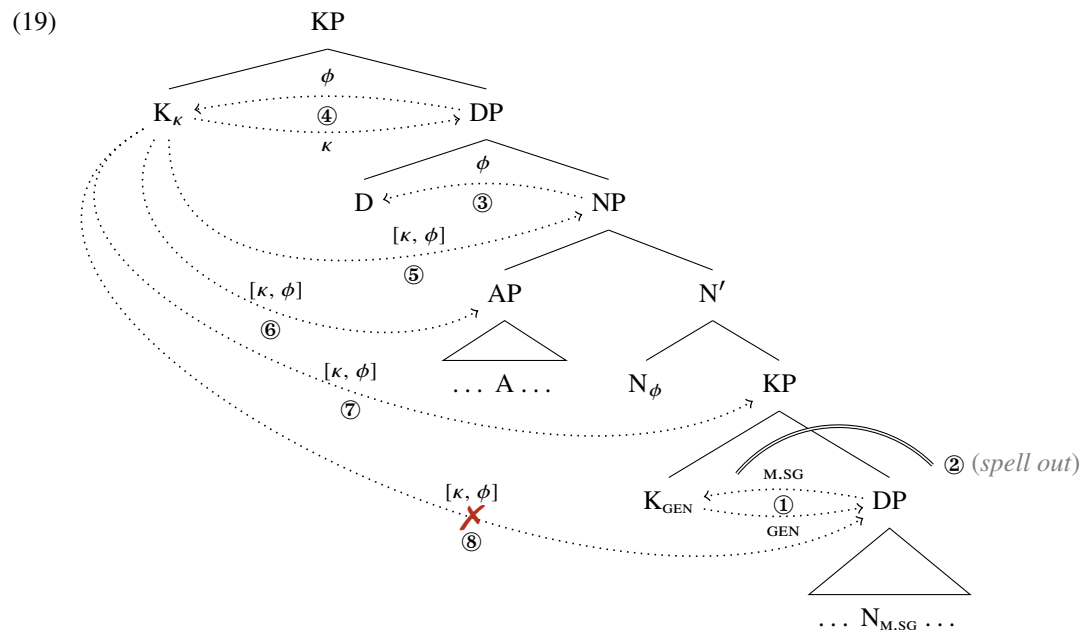
¹⁴By contrast, these – and more – typological generalizations are derived in Bayırılı 2017 with the help of specific feature percolation principles and non-standard assumptions about phases.

¹⁵Similarly, given that gender-introducing heads are closer to N than number-introducing heads (see, e.g., Saldana *et al.* 2025 and references therein), the possibility of concord in gender – to the exclusion of number and case – is predicted. This prediction seems to be correct about Shughni (a Pamir language spoken in Tajikistan, Afghanistan, Pakistan, and China) and Bangla (an Indo-Aryan language of Southeast Asia); see Bayırılı 2017: 31–33.

¹⁶See, e.g., Citko 2014: §4.3 and references therein on DP phases (on the assumption that DPs are maximal nominal projections), as well as, e.g., Richards 2010, Norris 2014, and Levin 2015, for explicit analyses of KPs (and PPs) as phases.

head is spelled out only when the next phase head is about to be merged. Moreover, the approach presented in this paper is compatible with the *horizons* approach to locality (Keine 2019, 2020); on this approach, the horizon of probes on K heads, beyond which such probes cannot operate, would be KP, in consonance with the generalization that, by default, the horizon of probes on X is XP (Keine 2020: 197–198).¹⁷

Given these almost universal assumptions, the structure of (18) is as shown in (19) (where κ = DAT, ϕ = SG.F), assuming for concreteness that Polish nominals may have DP layers and that *tej* ‘this’ in (18) is a D (although nothing hinges on these additional assumptions).



In the first interesting step, the lower K Agrees with the lower DP, exchanging case (here GEN) and ϕ features (M.SG), as explained in §3.3 above. Then, when the lower KP is fully constructed, the lower DP is spelled out and becomes inaccessible for any further operations. We make a simplifying assumption that this KP, spelled out as *roku* ‘year.M.SG.GEN’, is a complement of the higher N, but – again – absolutely nothing hinges on this assumption, as long as this KP is anywhere inside the higher KP. The following steps three to six are exactly as explained in §3.3 above. What is new here is that, in step seven, the feature bundle $[\kappa, \phi]$ is assigned to the lower – genitive – KP. Given that in Polish Ks are phonetically empty and there are no phrasal affixes that could expone features of KPs, this step has no visible effect. Finally, any further probing attempts, e.g., an attempt to probe the lower DP, fail, given that this DP and all its constituents are spelled out by this stage.

This analysis carries over to other familiar European languages, but it also predicts that, in languages that overtly expone features of K(P)s, multiple cases resulting from concord could be exponed. This prediction is borne out.

In this context, consider Lardil, which is a concord language, as shown in (20) (Richards 2013: 46, (12)).

- (20) Kara nyingki kurri [kiin-i mutha-n thungal-i]? (LBZ)
 Q you see that-ACC big-ACC tree-ACC
 ‘Do you see [that big tree]?’

It also, as many other Australian languages, displays overt case stacking (Richards 2013: 50, (23)):

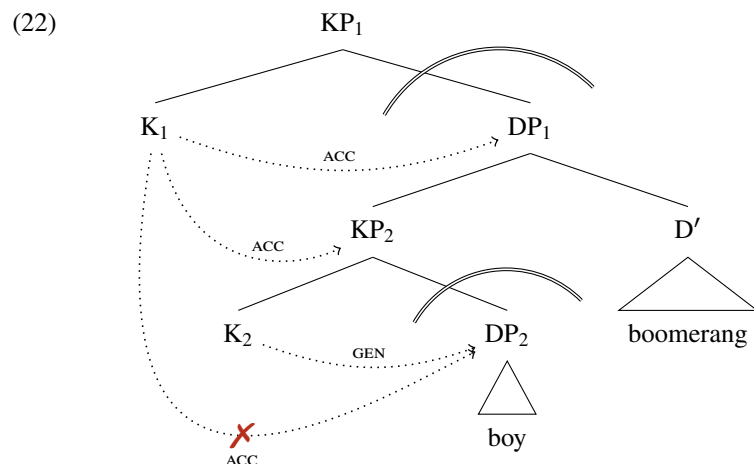
- (21) Ngada derlde [[marun-ngan-i] wangalk-i]. (LBZ)
 I.NOM break boy-GEN-ACC boomerang-ACC
 ‘I broke [[the boy’s] boomerang].’

In (21), the possessor within the direct object bears both a GEN affix, as possessors do, and an ACC affix, as a constituent of the direct object. This is directly predicted by the proposed analysis of concord, on the assumption that, in Lardil, case can be exponed as a suffix of any constituent, including a KP.¹⁸ To see this, consider the structure of the relevant KP,

¹⁷Keine (2019, 2020) argues that horizons are compatible with – and independently needed of – phases. This is true also in the context of the current account, where the KP horizon of probes on K heads obviates the need for an independent explanation of why one KP within another KP cannot move to that larger KP’s Spec escape hatch and be assigned an additional case there; cf. fn. 21.

¹⁸See Pesetsky 2013: §9.3 for a slightly different suggestion, which inspired the analysis presented here.

presented in (22), and note that KP_2 bears two cases: GEN assigned by K_2 and ACC assigned by K_1 , in this order. Hence, this KP_2 is expounded as *marun-ngan-i* ‘boy-GEN-ACC’.¹⁹

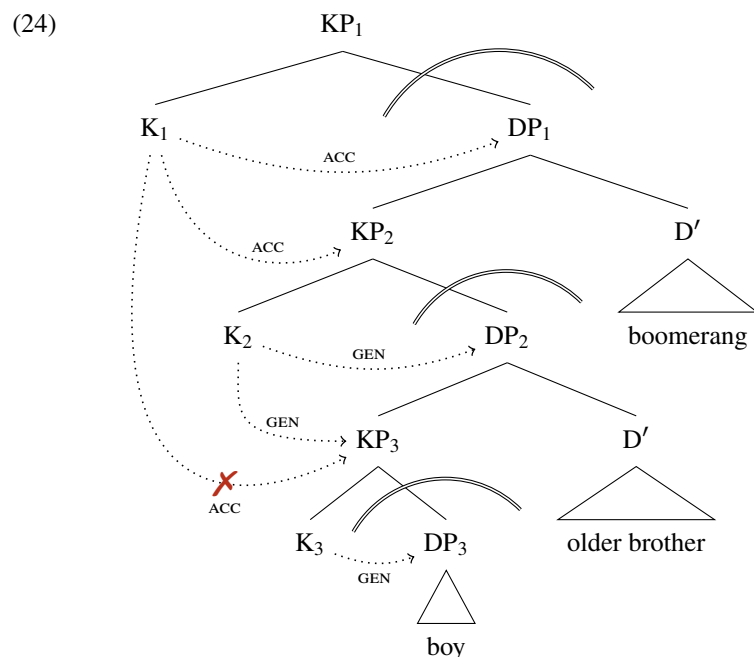


Note that the higher K_1 is not able to flag DP_2 or anything within it, as this DP_2 is spelled out by the time K_1 merges. This predicts that only KP_2 as a whole may be marked by ACC, but nothing else within it may. This prediction is again borne out.

It might at first seem baffling that, in Lardil, doubly embedded possessors within an ACC nominal phrase do not bear an ACC suffix. This is illustrated in (23) (Richards 2013: 50, (24)), where *marun-ngan* ‘boy’ is only marked for GEN, instead of GEN-GEN-ACC or – given the widespread rule of haplology of affixes of the same type (cf. fn. 11) – at least GEN-ACC.²⁰

- (23) Ngada derlde [[[marun-ngan] thabuji-kan-i] wangalk-i]. (LBZ)
 I.NOM break boy-GEN older.brother-GEN-ACC boomerang-ACC
 ‘I broke [[[the boy’s] older brother’s] boomerang].’

However, this immediately follows from the current analysis. Consider the tree in (24), corresponding to (23), with only the most important effects of Agree indicated.



The initially baffling GEN form in (23) corresponds to KP_3 in (24). The head of KP_3 is intrinsically GEN, and KP_3 is also flagged as GEN by the higher K_2 , but GEN is realized just once, for reasons alluded to above. What is crucial is that KP_3

¹⁹In order to avoid excessive clutter, only case flagging is shown in (22). For concreteness, the possessor is assumed to reside in Spec,DP (whether it is base-generated there or moved from within the embedded NP), but nothing hinges on this assumption.

²⁰This fact is noted but not explained in Richards 2013.

cannot be flagged with ACC. This is because K_2 is a phase head, so its DP_2 complement – and the KP_3 within it – is spelled out before the accusative K_1 has a chance to flag KP_3 with ACC. Hence, on the proposed approach to concord, no additional assumptions regarding feature percolation are needed to account for the puzzling pattern of case stacking on singly embedded possessors and the lack of case stacking on doubly embedded possessors.²¹

In summary, the proposed account of concord as insatiable Agree is compatible with a wide range of assumptions about the make up of nominal structures and with various implementations of locality constraints (phases, horizons), as long as the standard assumptions are adopted that maximal nominal projections are phases and/or that XPs are horizons for probes on X heads. On these standard assumptions, the proposed account correctly predicts that maximal nominal projections *within* larger nominal phrases often do not participate in concord, as in Polish (where features on $K(P)$ are not independently expounded), or do so only in a limited way, resulting in up to two stacked cases, as in Lardil (where features on KP are expounded directly).

It remains to be seen whether this analysis extends to all other languages with case stacking resulting from concord, but it seems to extend to at least some, given the observed “resistance to sequences of more than two case markers, as shown by some languages in Suffixaufnahme constructions” (Moravcsik 1995: 457) and given that some of the “case markers” in languages that seem to allow for more than two such markers seem to be assigned by T or C, rather than by K.²² Note however that, in some languages, such as Korean, there is another source of overt case stacking, namely, movement of the nominal phrase from one cased position to another cased position. In such languages, if there are constructions involving movement through more than two case positions, more than two case markers may be stacked, as in raising-to-object constructions in Amis (Austronesian, Taiwan), illustrated with (25) (Chen 2018: 21, (14)).²³

- (25) Ma-fana’ kako to-ko-ni Panay mi-liyas-to Inacila. (AMI)
 IPFV.STAT-know 1SG.NOM ACC-NOM-GEN Panay IPFV.AV-leave-ASP yesterday
 ‘I know that it was Panay who left yesterday.’

The analysis in Chen 2018 may be recast in the following terms. First, subjects of imperfective clauses are GEN, but when they are contrastive topics, as in (25), they move to a NOM position. Additionally, when the resulting clause is a complement of a raising-to-object verb, again as in (25), the subject moves to an ACC position, resulting in the three case markers in *to-ko-ni Panay* ‘ACC-NOM-GEN Panay’ (where *Panay* is a proper name). Such instances of potentially more than two stacked cases must be explained independently of any particular analysis of concord.²⁴

3.6 Interim Summary

The theoretical advantage of the analysis of concord proposed in this section over previous approaches (e.g., Norris 2014 and Bayırlı 2017) is that it does not posit any special feature percolation principles and does not rely on any non-standard locality assumptions. Instead, it relies on relatively recent but already widely adopted approaches to Agree, on which a single probe may target multiple goals, flagging each with its features: case and ϕ features in many languages, or just ϕ features in some. Empirically, this approach predicts out of the box not only the most standard concord facts, but also certain typological generalizations (e.g., that in languages with overt case and ϕ features pure ϕ concord is possible but pure case concord is not) and certain initially baffling case stacking facts.

In the following sections, this approach is stress tested by applying it to heterogeneous case and ϕ patterns in Slavic and Finnic.

²¹ While no additional assumptions regarding feature percolation are needed, it needs to be explained why KP_3 in (24) cannot receive ACC from K_1 by moving to the Spec, KP_2 “escape hatch”. While there are various ways to prevent that (e.g., an appropriate version of an anti-locality constraint), we assume here for concreteness that probes on K heads have KP horizons, in accordance with the generalization about default horizons in Keine 2019, 2020. If so, such a movement to Spec, KP_2 would not result in ACC flagging, as KP_2 would still be a horizon for the K_1 probe.

²² This seems to be the case in Kayardild, an Australian language related to Lardil, where four such markers are reported in the following example (Evans 1995: 406, (35a)):

- (i) Ngada mungurru, [maku-ntha yalawu-jarra-ntha yakuri-naa-ntha thabuju-karra-nguni-naa-ntha mijil-nguni-naa-nth]. (GYD)
 1SG know woman-C.OBL catch-PST-C.OBL fish-M.ABL-C.OBL brother-GEN-INS-M.ABL-C.OBL net-INS-M.ABL-C.OBL
 ‘I know that the woman caught the fish with brother’s net.’

Note, however, that out of the 4 suffixes in *thabuju-karra-nguni-naa-ntha* ‘brother-GEN-INS-M.ABL-C.OBL’, only *karra* ‘GEN’ and *nguni* ‘INS’ are true case markers. By contrast, the “modal ablative” *naa* ‘M.ABL’ marks the past tense and, hence, might be the result of flagging by T, while the “complementizing case” *ntha* ‘C.OBL’ marks all constituents of the subordinate clause, including verbal constituents, so it may be analyzed as the result of flagging by (covert) C. Adopting the horizons theory of Keine 2019, 2020, KPs would not be horizons for such higher heads, so the possibility of such additional markers even on the embedded genitive KP is expected on the current approach.

²³ ASP = aspectual marker, STAT = stative, other abbreviations follow Leipzig Glossing Rules.

²⁴ Such an analysis is well outside the scope of the current paper, but one possibility is that it involves Parallel Merge (Citko 2000, 2005, 2011, Citko and Gračanin-Yüksek 2020) of KPs with additional K heads. In the case of (25), the initial KP_{GEN} might be Parallel Merged with K_{NOM} when it moves to the relevant contrastive topic position, and then the resulting KP_{NOM} could be Parallel Merged with K_{ACC} on its way to the accusative object position (cf. Chen 2018: ch. 4). On such an analysis, *to-ko-ni* ‘ACC-NOM-GEN’ is the direct exponence of the three K heads, in the predicted order, as opposed to the exponence of up to two cases on a single KP in Lardil.

4 Heterogeneous Patterns in Slavic

4.1 Introduction

A well-known feature of north (i.e., East and West) Slavic numeral–noun constructions (NNCs) is the contrast between their heterogeneous case patterns in direct cases (NOM, ACC) and homogeneous patterns in oblique cases (GEN, DAT, LOC, INS) – see, e.g., Mel’čuk 1985 and Babby 1987 on Russian, and Franks 1994, 1995 on Slavic more generally. This is illustrated on the basis of Polish data in (26).

(26)	position	‘good.A’	‘books.N’	‘five.Num’	‘books.N’	‘mass.N’	‘books.N’
	NOM	<i>dobrze.NOM</i>	<i>książki.NOM</i>	<i>pięć.NOM</i>	<i>książek.GEN</i>	<i>masa.NOM</i>	<i>książek.GEN</i>
	ACC	<i>dobrze.ACC</i>	<i>książki.ACC</i>	<i>pięć.ACC</i>	<i>książek.GEN</i>	<i>masę.ACC</i>	<i>książek.GEN</i>
	GEN	<i>dobrych.GEN</i>	<i>książek.GEN</i>	<i>pięciu.GEN</i>	<i>książek.GEN</i>	<i>masy.GEN</i>	<i>książek.GEN</i>
	DAT	<i>dobrym.DAT</i>	<i>książkom.DAT</i>	<i>pięciu.DAT</i>	<i>książkom.DAT</i>	<i>masie.DAT</i>	<i>książek.GEN</i>
	LOC	<i>dobrych.LOC</i>	<i>książkach.LOC</i>	<i>pięciu.LOC</i>	<i>książkach.LOC</i>	<i>masie.LOC</i>	<i>książek.GEN</i>
	INS	<i>dobrymi.INS</i>	<i>książkami.INS</i>	<i>pięcioma.INS</i>	<i>książkami.INS</i>	<i>masą.INS</i>	<i>książek.GEN</i>

The first two columns contain Polish for ‘good books’ in various case positions.²⁵ As expected of a full concord language, the adjective agrees with the noun in case, number and gender. The final two columns contain Polish for ‘a mass of books’, headed by the singular noun *masa* ‘mass’. Again, as expected, the nominal complement of this noun is in the genitive in all case positions. By contrast, the Polish numeral–noun construction for ‘five books’ in the middle columns (in bold) behaves like the noun–noun construction in NOM and ACC, where the noun is in the genitive, and like the adjective–noun construction in the other positions, where the noun agrees with the numeral in case.^{26,27} Similar patterns of NNCs are observed in other West Slavic languages (including Czech) and in East Slavic (including Russian and Ukrainian), but not in South Slavic.

4.2 Analysis

4.2.1 Idea

The following account of this basic pattern of north Slavic NNCs is based on that of Pesetsky 2013 (which in turn was based on Richards 2013), but it is free of the considerable number of controversial assumptions made there and it attempts to be more explicit and precise.

The basic idea of Richards 2013 and Pesetsky 2013 is that Russian is a case stacking language, but only the last assigned – i.e., the outermost – case is overtly realized:²⁸

(27) *Expone Outermost*: Expone only the outermost feature bundle. (cf. Pesetsky 2013: 11, (7))

In oblique positions, externally assigned case percolates down and ends up on nominals previously marked as GEN, resulting in case stacking. But given the *Expone Outermost* rule in (27), only the oblique case is expounded. This approach is equally applicable to Polish and it accounts for the GEN, DAT, LOC, and INS rows in the table in (26). However, if nothing more is said, the GEN case on the noun in the NOM and ACC rows is unexpected: it should be NOM and ACC, respectively. In order to prevent the spread of NOM and ACC to the genitive noun, a relatively large number of non-standard assumptions are made in Pesetsky 2013.²⁹

We modify this general covert case stacking idea by following the insight in Grishin 2024 that there are phenomena which call for a different exponence rule than (27), namely, a rule like (28):³⁰

(28) *Expone Most Marked*: Expone only the most marked feature bundle.

²⁵ Apart from these 6 cases, Polish also has the vocative case used for address and not selected by any predicates. As such, it may be regarded as being outside the case system (cf. Blake 2001: §1.2.3) and will be ignored in the rest of this paper, as is common in discussions of Slavic numerals. (Vocative NNCs always have the same form as nominative in Polish, as vocative always equals nominative in the plural and NNCs are always plural.)

²⁶ In the genitive position, it is not immediately clear whether GEN on the noun is assigned as in nominative and accusative positions, or whether it agrees with the numeral as in other oblique positions. Russian paucal numerals (§5) and numeral constructions in Finnic (§6) show that GEN patterns with other oblique cases there, so we assume it also does in Polish, as described in the main text.

²⁷ Note that the numeral in the NOM row is marked in gray, as there are good arguments for analyzing Polish numeral subjects as actually in the accusative (see, e.g., Schenker 1971, Franks 1994, 1995, Przepiórkowski 1996, 1999, 2004, Miechowicz-Mathiasen 2012, Willim 2015, Witkoś *et al.* 2018, Citko *et al.* 2018). The question of the source of this accusative is outside the scope of this paper.

²⁸ This rule was renamed from the original *One-Suffix Rule* for added precision and generalized for compatibility with our set of assumptions. While in Pesetsky 2013 it is only applied to stacked cases in Russian, Grishin 2024 extends it to stacked ϕ -feature bundles in Passamaquoddy-Wolastoqey (Eastern Algonquian; Canada and USA).

²⁹ These include some very special properties of null Determiners resulting in their selective assignment of the nominative case, an analysis of those accusative objects which are (commonly taken to be) syncretic with nominative as *actually* NOM, some very special structures of prepositional phrases (apparently) headed by prepositions assigning ACC or GEN, etc.

³⁰ A similar idea applied to Russian cardinals, which foreshadows the analysis proposed below, may also be found in Matushansky 2008: §6 (a proceedings paper we became aware of late in the process of writing the current article).

Such a rule seems to be at play in Gothic free relatives, as shown in (29) (Harbert 1992: 111 apud Grishin 2024: 47):³¹

- (29) a. [Pan=ei frijos] siuks ist. (GOT)
 who.ACC=C you.love_{ACC} sick is_{NOM}
 ‘(The one) whom you love is sick.’
 b. [Po=ei ist us Laudeikaion] jus ussiggwaid. (GOT)
 which.ACC=C is_{NOM} out Laodicea you read_{ACC}
 ‘And read (the one) which is from Laodicea.’

Unlike in contemporary German and many other languages, where the relative pronoun must simultaneously satisfy case requirements of both the matrix and the embedded clause, in Gothic the relative pronoun occurs in the more marked case of the two required in the two clauses. For example, in (29a), where ACC is expected in the relative clause and NOM in the matrix clause, the pronoun occurs in ACC. By contrast, in (29b) it is the matrix clause that expects ACC, with the relative clause expecting NOM, but the pronoun is still in ACC.

This is expected if the *Expone Most Marked* rule in (28) is at play, on the assumption that ACC is more marked than NOM. But this assumption is independently needed, as argued at length – on the basis of a number of phenomena, including case syncretisms – in Caha 2009, where the following hierarchy of morphological cases is proposed (with the least marked cases on the left):

- (30) NOM < ACC < GEN < DAT < INS < . . .

Given this independently motivated case hierarchy, the *Expone Most Marked* rule immediately explains the puzzling case patterns in Slavic NNCs, as shown in (31).³²

(31)

position	‘five.Num’	‘books.N’
NOM	<i>pięć.NOM</i>	<i>książek.GEN.NOM</i>
ACC	<i>pięć.ACC</i>	<i>książek.GEN.ACC</i>
GEN	<i>pięciu.GEN</i>	<i>książek.GEN.GEN</i>
DAT	<i>pięciu.DAT</i>	<i>książkom.GEN.DAT</i>
LOC	<i>pięciu.LOC</i>	<i>książkach.GEN.LOC</i>
INS	<i>pięcioma.INS</i>	<i>książkami.GEN.INS</i>

This table repeats the middle columns of the table in (26), but with case stacking shown explicitly, and with the most marked case in each stack boxed. It is exactly the boxed cases that are expounded, as predicted by the *Expone Most Marked* rule in (28).³³

4.2.2 On Structures of NNCs

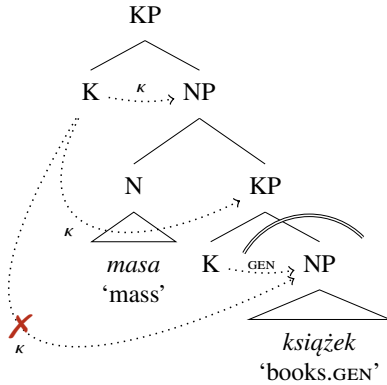
Despite the vast literature on Slavic numeral–noun constructions, there is still no agreement on almost any aspect of their syntax. However, the one aspect that is crucial for the analysis below is currently generally accepted: numerals occur within an extended nominal projection, rather than introducing a separate nominal projection. In our terms, this means that, while noun–noun constructions involve two KPs, ultimately headed by the two nouns, numeral–noun constructions involve a single KP ultimately headed by the single noun. This is schematically shown in (32)–(34) (for perspicuity, without DP or any other layers).

³¹A subscripted case value indicates case expectations of a given word (or an associated functional head).

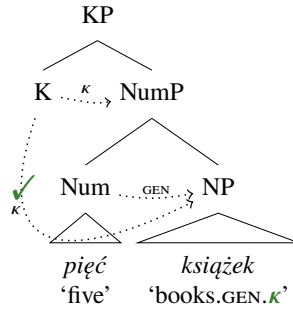
³²Recall (from fn. 27) that, in Polish, numeral subjects are actually ACC, not NOM – hence the gray NOM in (31). However, in Russian and other Slavic languages, such numeral subjects may occur in the nominative, and the argument in the text is valid whether the case assigned to numeral subjects is NOM or ACC, as both are less marked than GEN.

³³In the GEN position, there are two GEN feature bundles on the noun. In §5 we will see that it is the outermost of these that is expounded.

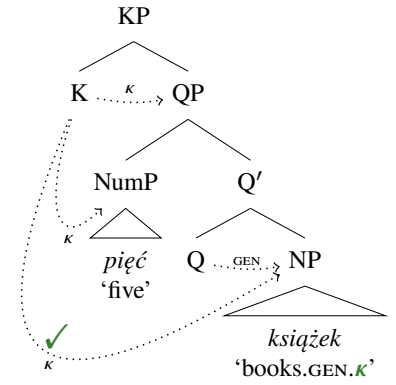
(32) noun–noun construction:



(33) NNC (Num is head):



(34) NNC (NumP is in Spec,QP):



In (32), corresponding to *masa książek* ‘(a) mass (of) books’, the lower NP is immune to the probe on the higher K, as it is already spelled out when that probe becomes active, as discussed in §3.5 above. So, whatever case the higher K bears, it only affects the higher noun, *masa* ‘mass’, and not the lower noun, *książek* ‘books.GEN’.

The trees in (33)–(34), by contrast, correspond to the two most common approaches to Slavic numeral–noun constructions. On the first, illustrated in (33), numerals are heads in extended nominal projections, and they directly assign the genitive case to their nominal complements. This kind of structure is assumed for Polish in Klockmann 2017 and – for various Slavic languages – in countless works before.³⁴ On the second approach, illustrated in (34), numerals are specifiers of empty quantificational heads in extended nominal projections, and it is such empty heads, Q, that are responsible for the genitive on the noun. Such structures are assumed for Polish in Witkoś *et al.* 2018 and – for various Slavic languages – in multiple earlier works.

As shown in (33)–(34), whichever of the two popular approaches to the structure of NNCs is adopted, case stacking is predicted, and for the same reason: there is no phase boundary between the K and the genitive NP that would block flagging the NP with κ . Hence, the noun ends up with two cases, GEN and κ , and expones GEN when κ is the less marked NOM or ACC; otherwise, it expones the κ , as the most marked case on the stack. This analysis remains valid when more layers in extended nominal projections are assumed, e.g., also DP and/or NumberP, as long as these additional layers do not constitute additional phase boundaries.

4.3 Additional Confirmation

The aim of this section is to present two idiosyncratic phenomena which confirm the analysis above. In particular, they show that the heterogeneous vs. homogeneous case pattern in Slavic numeral–noun constructions is not some quirk of NNCs and is not limited to case, but rather follows from the general mechanisms of concord in §3. We start with something interesting.

4.3.1 Something Interesting

There are a handful of Polish pronouns, all containing the root *c(o)*, which display a heterogeneous vs. homogeneous case pattern similar to that of numerals; these are: *co* ‘what’, *coś* ‘something’, *cokolwiek* (and its colloquial/archaic variant *cośkolwiek*) ‘whatever’, and *nic* ‘nothing’. This case pattern is illustrated in (35) (the bolded initial columns) on the basis of *coś ciekawego* ‘something interesting’. Note that, while in NNCs the numeral occurs in the expected case and the noun displays the mixed case pattern, here the pronoun occurs in the expected case and it is the adjective that displays the mixed pattern.

(35)	position	‘something interesting’	‘somebody interesting’
	NOM	coś.NOM <i>ciekawego.GEN</i>	<i>ktoś.NOM</i> ciekawym.NOM
	ACC	coś.ACC <i>ciekawego.GEN</i>	<i>kogoś.ACC</i> ciekawego.ACC
	GEN	czegoś.GEN <i>ciekawego.GEN</i>	<i>kogoś.GEN</i> ciekawego.GEN
	DAT	czemuś.DAT <i>ciekawemu.DAT</i>	<i>komuś.DAT</i> ciekawemu.DAT
	LOC	czymś.LOC <i>ciekawym.LOC</i>	<i>kimś.LOC</i> ciekawym.LOC
	INS	czymś.INS <i>ciekawym.INS</i>	<i>kimś.INS</i> ciekawym.INS

³⁴However, in Klockmann 2017, it is not Num directly but rather dependent case mechanisms (not made precise there) that are responsible for the genitive on the noun.

This handful of pronouns are the sole exception to the rule that adjectives within nominal phrases always agree in case with the nominal heads in Polish; as shown in (35), in NOM and ACC positions adjectives modifying *coś* ‘something’ occur in the genitive, instead of the expected nominative or accusative (which both would have the form *ciekawego*). In other case positions, the adjective occurs in the expected agreeing case, mirroring the distribution of nouns in NNCs.³⁵ This behavior should be contrasted with the usual situation, exemplified in the final columns of (35), which illustrate that pronouns based on the root *kt(o)*, e.g., *ktoś* ‘somebody’, do not display this case pattern; here the adjective always agrees with the pronoun, as it does elsewhere in Polish.

We do not attempt to explain why exactly *coś* ‘something’, but not *ktoś* ‘somebody’, flags its adjectival modifier with GEN.³⁶ But, given that it does, the pattern shown in (35) is fully expected on the approach to concord presented in §3: since AP is not a phase, the higher K may flag this AP with case κ , on top of GEN received from *coś*, resulting in case stacking and the realization of the most marked case. On this account, the only idiosyncrasy of the Polish pronominal root *c(o)* is that it flags the adjective with GEN; the mixed pattern seen in (35) follows from the lack of phase boundary between the higher K and the AP, and from the *Expone Most Marked* rule in (28).

This is confirmed by the fact that a genitive KP dependent of *coś* ‘something’ must stay in the genitive, regardless of the external case. While such an additional genitive nominal dependent of *c(o)* is often awkward due to the structural ambiguity it may create,³⁷ it is relatively easy to find occurrences in corpora that exemplify this pattern, e.g., (36) from the National Corpus of Polish. (POL)

- (36) Chciałbym dla odmiany przeczytać coś lżejszego Twojego autorstwa. . .
 want.M.1SG.SBJV for change read.INF something.ACC lighter.GEN your.GEN authorship.GEN
 ‘I’d like to read something lighter by you, for a change.’
 (lit.: ‘... something.ACC lighter.GEN [your authorship].GEN’)

In (36) the *coś* phrase is in an ACC position, so not only the KP *Twojego autorstwa* ‘your authorship’, but also the AP *lżejszego* ‘lighter’ is in GEN. By contrast, when such a phrase occurs in an oblique (e.g., LOC) position, *coś* ‘something’ and the AP occur in the oblique case, as discussed above, by the KP remains in GEN, as shown in the constructed (37).

- (37) Porozmawiajmy o czymś lżejszym Twojego autorstwa. (POL)
 talk.1PL.IMP about something.LOC lighter.LOC your.GEN authorship.GEN
 ‘Let’s talk about something lighter by you.’
 (lit.: ‘... something.LOC lighter.LOC [your authorship].GEN’)

The structure of such nominal phrases is schematically given in (38), without commitment to any particular position of the AP and the inner KP; what is crucial is only that the inner KP is a phase, so the NP within it is not accessible to the probe on the higher K, while there is no similar phase boundary between the higher K and the AP.

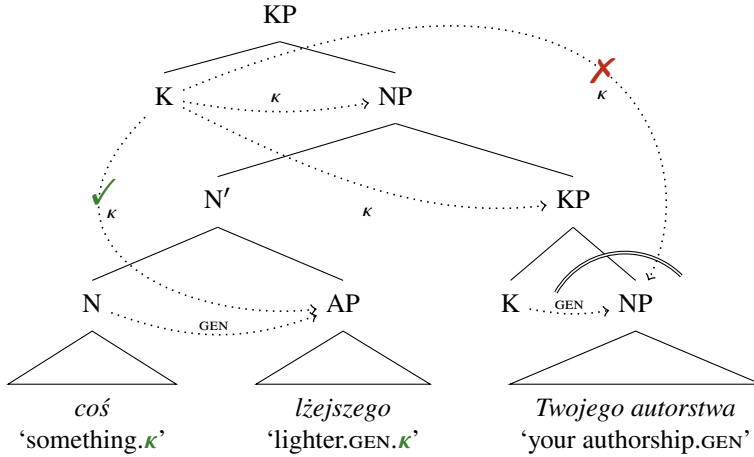
In summary, the ‘something interesting’ idiosyncrasy confirms the general approach to concord and the particular analysis of Slavic numeral–noun constructions offered above. There is nothing special about numeral constructions that gives rise to the famous heterogeneous vs. homogeneous case pattern: this pattern follows from the usual assumptions about nominal phases (only maximal nominal projections – here, KPs – are phases) and the *Expone Most Marked* rule. As independently cased nominal projections within other nominal projections are usually KPs, which are phases, no case stacking is observed. However, case stacking and the resulting mixed case exponence pattern is predicted whenever some non-phasal constituent of a KP receives an independent case. A conspicuous and productive construction where this happens is north Slavic NNC, where numerals (assuming the analysis in (33)) or empty Q heads (assuming (34)) flag their NP complements with GEN, but also – idiosyncratically – pronouns based on *c(o)*, which similarly flag their AP dependent. In both cases, the flagged constituent is not a phase, which leads to case stacking and the observed heterogeneous vs. homogeneous case pattern.

³⁵Some Lithuanian pronouns, including *kažkas* ‘something’, behave similarly, even though Lithuanian NNCs do not display the Slavic heterogeneous vs. homogeneous case pattern (Rutkowski and Szczegot 2001).

³⁶In terms of Klockmann 2017, the root *c(o)* – but perhaps not *kt(o)*, etc. – would be analyzed as idiosyncratically semi-lexical, in the sense that it hosts certain functional features (here; a GEN-flagging probe); see also Rutkowski and Szczegot 2001 for a similar analysis.

³⁷For example, *coś ciekawego Szekspira* ‘something interesting.GEN Shakespeare.GEN’ may be understood as ‘something interesting/curious by Shakespeare’, or as ‘something by the curious/interesting Shakespeare’.

(38)



4.3.2 Post-Prepositional Forms of 3rd Person Pronouns

There is another little known phenomenon that has been considered an inexplicable “idiosyncrasy at the crossroads of idiosyncrasies” (Świdziński and Derwojedowa 2004). One “idiosyncrasy” of Polish and other north Slavic languages is the heterogeneous vs. homogeneous case pattern of NNCs discussed in §§4.1–4.2. Another “idiosyncrasy” present in many north Slavic languages are the specific post-prepositional forms of 3rd person personal pronouns (3PPPs). This is exemplified in the constructed minimal pair in (39).³⁸

- (39) a. Zaprosiliśmy **ich**. b. Czekaliśmy na **nich**. (POL)
 invited_{ACC.F.1PL M.3PL.ACC} waited_{F.1PL ON_{ACC} M.3PL.ACC.@P}
 ‘We invited them.’ ‘We waited for them.’

In (39a), the regular form of the accusative M.3PL personal pronoun occurs, *ich*. By contrast, in (39b), the “post-prepositional” accusative form of the pronoun – here marked as @P – must be used, i.e., *nich*. Neither form may be used in the other context. In Polish, such special @P forms occur in post-prepositional ACC, GEN, and DAT positions,³⁹ while in other Slavic languages that make the same distinction, including Russian, also INS is similarly affected.⁴⁰ Philippova (2018) convincingly argues that, in Russian, @P forms are the result of P’s category flagging and that the initial *n-* expones the P category on the pronoun. We adopt this analysis to Polish, noting its resemblance to Clem and Deal’s (2024) analysis, which assumes that all flagging involves category flagging (see §2 above).⁴¹

Now, the “idiosyncrasy at the crossroads of (the two) idiosyncrasies” presented above – at the intersection of mixed case pattern in NNCs and post-prepositional forms of 3PPPs – is illustrated by the following (constructed) minimal pair:

- (40) a. Czekaliśmy na **ich** propozycje. (POL)
 waited_{F.1PL ON_{ACC} 3PL.GEN suggestions.F.PL.ACC}
 ‘We were waiting for their suggestions.’ (lit.: ‘... for they.GEN suggestions.ACC’)
 b. Czekaliśmy na **nich** {pięciu / pięć}. (POL)
 waited_{F.1PL ON_{ACC} 3PL.GEN.@P five.M/F.ACC}
 ‘We were waiting for the five of them.’ (lit.: ‘... for they.GEN.@P five.ACC’)

³⁸In both sentences, the subject is a *pro*-dropped 1PL.F, as evidenced by agreement features on the verb.

³⁹This distinction does not show either in LOC, for the simple reason that LOC is a solely post-prepositional case, so all relevant pronominal forms start in *n-*, or in NOM, probably because of the different phonological makeup of relevant NOM pronouns; for example: *oni/one* ‘M/F.3PL.NOM’ vs. *ich* ‘3PL.ACC/GEN’ and *im* ‘3PL.DAT’. See <http://sgjp.pl/leksemy/#90012/on> (Woliński *et al.* 2020) for the full paradigm of Polish 3PPPs (in Polish), or Swan 2002: 156–157 for a description in English.

⁴⁰In INS, Polish behaves like many South Slavic languages: all forms start in *n-*.

⁴¹Interestingly, a somewhat similar phenomenon is observed in languages as different from Slavic as Nez Perce (Penutian; USA) and Shipibo (Panoan; Peru and Brazil). In both of these languages, cases traditionally called ergative and genitive are only distinguished by personal pronouns: 1st/2nd person pronouns in Nez Perce (Deal 2010: 78, fn. 9) and 1st/3rd person singular pronouns in Shipibo (Baker 2015: 20–21, 166–171). Baker (2015) analyzes such facts by claiming that ergative and genitive are the same (high dependent) single case, and that the different pronominal forms reflect an independent ±NOMINAL featural distinction. Translating this analysis into our terms, genitive in Nez Perce and Shipibo is simply ergative, but additionally marked with the nominal category of the selecting item (N or D), with some pronouns being the only items sensitive to this category flagging.

This contrast – especially, the @P form *nich* in (40b) – is surprising at first. This is because, according to the otherwise relatively robust “morphosyntactic locus criterion” criterion of headedness (see, e.g., Zwicky 1985 and Hudson 1987), both accusative nominal phrases in (40a–b) seem to be headed by the accusative forms: *propozycje* ‘suggestions’ and *pięciu / pięć* ‘five.M/F’, respectively: it is these forms that bear the case expected by the selecting preposition *na*, while the pronouns occur in the genitive case. Hence, on this view, the genitive pronouns are not direct complements of the preposition, but rather are dependents *embedded within* such direct complements. And yet, the one in (40a) must be realized as *ich*, and the one in (40b) is the @P form *nich*.⁴²

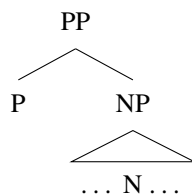
This initially puzzling contrast – requiring very specialized feature percolation rules on an earlier account of Świdziński and Derwojedowa 2004 – is predicted on the current analysis without any non-standard assumptions (but see below for some standard assumptions about phases in PPs). In (40a), the genitive pronoun is within the usual genitive KP complement of the noun, i.e., a phase. For this reason, it cannot be flagged by P – either with case or with P’s category. By contrast, the genitive pronoun in (40b) is a part of a single extended nominal projection, i.e., it is not within a phase. For this reason, it *can* be flagged by P – both with case and with P’s category. If this analysis is on the right track, then the genitive in (40a) is the only case feature borne by the pronoun, while the genitive in (40b) is the result of expounding the most marked case of the two cases on its stack: GEN (assigned by the numeral or Q) and ACC (assigned by P). Hence, a prediction of this analysis is that, if a case more marked than GEN is assigned by the preposition, it should be expounded together with @P, but not when the pronoun has the usual non-@P form. The following constructed minimal pair shows that this prediction is borne out:

- (41) a. Głosowałyśmy przeciw **ich** propozycjom. (POL)
voted.F.1PL against_{DAT} 3PL.GEN suggestions.F.PL.DAT
‘We voted against their suggestions.’ (lit.: ‘... against they.GEN suggestions.DAT’)
- b. Głosowałyśmy przeciw **nim** pięciu. (POL)
voted.F.1PL against_{DAT} 3PL.DAT.@P five.M/F.DAT
‘We voted against the five of them.’ (lit.: ‘... against they.DAT.@P five.M/F.DAT’)

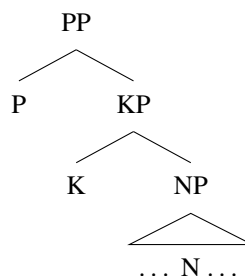
In (41a), the pronoun is protected by the KP phase, so it is flagged neither with P’s category, nor with the dative case assigned by the preposition. By contrast, the unprotected pronoun in (41b) is flagged with both, resulting in the DAT.@P form *nim*.⁴³ This shows that it is not a special feature of case flagging that is responsible for the mixed pattern in Slavic NNCs: other kinds of flagging, e.g., category flagging, are similarly sensitive to phase boundaries.

Now, we mentioned above that no “non-standard assumptions” are needed, but the reasoning above *is* based on some additional assumptions regarding the structure of PPs, namely, it assumes that Ps may flag into their nominal complements. The particular structure tacitly assumed above, with P replacing K as a case-flagger, is schematically shown in (42).⁴⁴

- (42) PP instead of KP:



- (43) PP on top of KP:



On this view, K and P form a natural class, but where K is a pure functional category, P is a semi-lexical category in the sense of Klockmann 2017: it introduces not only features and probes, but also a lexical root.⁴⁵ In particular, both K

⁴²In both (40a–b), the preposing of the pronoun within the nominal projection seems to be a strong tendency rather than a strict rule, as noted in Świdziński and Derwojedowa 2004 and further confirmed by attested examples such as the following:

(i) Do **obu nich** podbiega ewidentnie rozjuszony pies... (POL)
to_{GEN} both 3PL.GEN run.up obviously angry.NOM dog.NOM
‘An obviously angry dog runs up to both of them.’

⁴³As can be guessed, the non-@P form of the 3PL.DAT pronoun is *im*.

⁴⁴This tree is schematic in the sense that additional nominal projections may also occur there, e.g., DP and NumberP, as long as they are not phases; and similarly for (43).

⁴⁵See, e.g., Řezáč 2008 on similarities between PPs and oblique maximal nominal projections. Note that the phenomenon discussed in this subsection provides an argument against Řezáč’s (2008) particular analysis of obliques as PPs headed by empty Ps, as on this analysis all oblique 3PPs should have post-prepositional forms in the relevant Slavic languages, while only those selected by true Ps do. However, as Řezáč (2008) notes himself, nothing hinges on the additional layer of obliques being a PP. We believe that the main points of his analysis can be translated into the current account, by

and P are phase heads, and both normally host an insatiable probe responsible for concord.

The above account is also compatible with the more pronounced structure in (43), but only on the assumption that KP is not a strong phase in such a configuration.⁴⁶ If it were, then the complement of K would be spelled out before P is merged, so it would be inaccessible to category flagging by P, contrary to the analysis of post-prepositional forms of 3PPs offered in Philippova 2018 and adopted above.⁴⁷ A possible analysis compatible with the larger structure in (43) could be based on the common assumption that PPs are nominal extended projections (e.g., Grimshaw 2000) and on the so-called “context-sensitive” approach to phases, on which only and all maximal extended projections of lexical categories are phases, whatever these maximal extended projections are in a given context (e.g., Bošković 2014). This way, when the maximal nominal projection is a KP, it is a phase, but when it is a PP, the PP – rather than the KP it immediately dominates – is a phase.⁴⁸

In summary, the proposed approach to concord immediately explains the “idiosyncrasy at the crossroads of idiosyncrasies” discussed in this section – whether PP and KP form a natural class but stand in complementary distribution (see the schematic structure in (42)), or whether PP provides another layer on top of KP and thus becomes the sole phase in this extended projection (see (43)). As could be expected, various kinds of flagging – with case or with category – are sensitive to the same phase boundaries. We will see in the next section that this is also true of ϕ features.

4.4 Interim Summary

On the proposed approach to concord, the famous Slavic heterogeneous vs. homogeneous pattern in Slavic numeral–noun constructions falls out from general and independently needed principles. It follows directly from insatiable Agree, which results in feature bundle stacking, and from the *Exponere Outermost* rule, needed independently to account for Gothic free relatives, among other phenomena. The only idiosyncrasy is that Slavic numerals – or the Q heads associated with them – flag their dependents with GEN, an idiosyncrasy long accepted in Slavic linguistics and shared with some indefinite pronouns.

No *sui generis* principles are also needed to account for the puzzling behavior of Russian paucal numerals, discussed in the following §5, but only on one of the two dominant analyses of Russian paucals, one that assumes a special paucal case. On the other hand, the more interesting analysis in terms of the paucal number requires some additional assumptions, which, however, seem also needed to account for heterogeneous concord patterns in Finnic, discussed in §6.

5 Paucal Numerals in Slavic

5.1 Paucals in Polish

In a number of Slavic languages, the syntax of paucal numerals – typically expressing ‘two’, ‘three’ and ‘four’ – is more complex than that of higher numerals (‘five’ and above). One aspect of this complexity observed in languages such as Polish and Russian is that the syntax of the paucal construction depends to some extent on grammatical gender.

In Polish, it makes sense to distinguish 5 genders (Mańczak 1956): feminine (F), neuter (N), masculine-inanimate (M3), masculine-animate (M2), and masculine-human (M1). The names of the three masculine genders indicate a semantic animacy distinction, but the correlation of M2 with animacy is very imperfect: there are many inanimate M2 nouns (including all masculine names of car makes and dances, e.g., *Fiat* and *walc* ‘waltz’), and even some M2 nouns with a feminine natural gender denotation (e.g., *babsztyl* ‘woman.DEROGATORY’). The syntax of higher numeral constructions does not depend on gender; as discussed in the previous section, the noun must occur in GEN when the NNC is in a NOM or ACC position, regardless of the noun’s gender.

The situation is different in the case of paucal numerals, as shown in (44)–(45). Here, only in M1 is the typical numeral syntax observed, with the noun in the genitive (see (44)). By contrast, “adjectival” syntax is possible in all genders (including M1), where both the numeral and the noun occur in the case specific to a given syntactic position, e.g., in NOM (see (45)).

- (44) *dwóch ochroniarzy* (POL)
 two.M1 guards.M1.PL.GEN
 ‘2 (male) guards’

postulating certain differences between properties of various K heads in various languages, similarly to his analysis in terms of variability of properties of particular P heads.

⁴⁶If KP is a weak phase, i.e., if Phrase Impenetrability Condition is understood as in Chomsky 2001, then – on the assumption that PPs are not phases – the complement of K is still not spelled out when the KP merges with P, so it can be flagged with P’s categorial features.

⁴⁷It might seem that an account could be offered in terms of the whole KP being flagged with P by the preposition, resulting in the exponence of this P feature on K as the initial *n-*. However, this account does not extend to the kind of data in (40b)–(41b), where the @P form is embedded deeper, especially given the possible word-order variation signaled in fn. 42 above.

⁴⁸The structure in (43) is also compatible with the horizons approach of Keine 2019, 2020, on the assumption that P does not host the relevant probe, but only selects a KP hosting such a probe.

- (45) a. dwie lampy (POL)
two.F.NOM lamp.F.PL.NOM
'2 lamps'
- b. dwa {okna / stoły / psy} (POL)
two.N/M3/M2.NOM window.N.PL.NOM table.M3.PL.NOM dog.M2.PL.NOM
'2 {windows / tables / dogs}'
- c. dwaj ochroniarze (POL)
two.M1.NOM guards.M1.PL.NOM
'2 (male) guards'

In terms of the analysis of the previous section, the main difference between higher numerals and paucal numerals is that the former – but typically not the latter – assign the genitive, which gets expounded as the more marked case in NOM and ACC positions.⁴⁹ By contrast, typical paucal numerals do not assign GEN, so there is no competition and the nominative case is expounded in (45).

A natural analysis of this contrast is in terms of slightly different probes on the relevant numerals, shown below with explicit information about flagging and selectional restrictions:⁵⁰

- (46) Polish higher numerals: [INT: ϕ , FLAG: GEN], SEL: N
- (47) Polish paucal numerals: [INT: ϕ , FLAG: GEN], SEL: N.M1
[INT: ϕ , FLAG: —], SEL: N

That is, Polish paucals are in principle ambiguous: they may have the usual GEN-flagging probe, but only when they select for M1 Ns, or they may have a non-GEN-flagging probe, without any gender restrictions.⁵¹ Note that the first probe is the same as in the case of higher numerals (only selectional restrictions are slightly different), while the second differs from it minimally, in giving up GEN flagging (but selectional restrictions remain the same).

As expected, the difference between these two probes becomes invisible in oblique cases, e.g., DAT: whether the M1 noun is assigned GEN first or not, DAT is expounded – either as the more marked case than GEN, or as the only case:

- (48) przeciw dwóm {ochroniarzom / *ochroniarzy} (POL)
against_{DAT} two.DAT guards.M1.PL. {DAT/*GEN}
'against 2 (male) guards'

In summary, Polish paucals do not require any substantial extensions either of the general approach to concord advocated here, or of the particular analysis of Polish NNCs in terms of the *Expone Most Marked* rule.

5.2 Paucals in Russian

The syntax of Russian paucal constructions also depends on gender, but there is an additional complexity: when the paucal phrase occurs in a nominative or accusative position, most ad-paucal nouns occur in what looks like *singular* genitive, e.g., *stula* in (49),⁵² instead of the expected plural genitive (as in the case of higher numerals in Polish and Russian or M1 paucal numerals in Polish) or plural nominative (as in the case of non-M1 paucal numerals in Polish),

- (49) dva {stula / rjadá / vyxodnyx} (RUS)
two.M.NOM/ACC chair.M.SG.GEN(?) row.M.SG?.GEN? exiting.M.PL.GEN(?)
'2 {chairs / rows / days off}'

Assuming the usual repertoire of Russian cases and numbers, the form *stula* 'chair' may only be analyzed as SG.GEN.

However, there are 5 masculine nouns that may occur ad-paucally in a form that is not witnessed anywhere else: *rjad* 'row', *sled* 'trace', *čas* 'hour', *šag* 'step' and *šar* 'sphere, ball'. When in an ad-paucal position, these nouns often display a stress pattern that is different from the SG.GEN stress, e.g., *rjadá* (with end stress) in (49) instead of the SG.GEN *rjáda* (with stem stress).⁵³

Apart from these 5 nouns, another problem for the generalization that Russian paucal numerals combine with SG.GEN complements is the fact that de-adjectival nouns, such as *vyxodnyx* 'days off' (lit. 'exiting') in (49), occur in what looks

⁴⁹We continue to ignore the fact that numerals which assign GEN in the subject position in Polish are themselves in the accusative (cf. fn. 27), from which certain subject–verb agreement facts follow, as this is a topic for a separate paper. Nevertheless, this is the reason for different forms of the M1 numerals in (44) (*dwóch*.ACC=GEN) and (45c) (*dwaj*.NOM).

⁵⁰We will say more about satisfaction conditions in §5.2; as long as attributive adjectives are not considered, either SAT: ϕ or SAT:– may be assumed.

⁵¹While positing ambiguity is always suspicious, we believe that it is real and reflects diachronic development of paucal numerals from purely adjectival forms; see diachronic comments on Russian paucals in §5.2. The process of numeralization creates ambiguities also at the other end of the scale, where words such as *tysiąc* 'thousand' and *milion* 'million' are ambiguous between numerals and nouns in contemporary Polish (see, e.g., Przepiórkowski 1999: 194–196 and references therein).

⁵²The full paradigm of *dva stula* 'two chairs' is given in the table in (53) below.

⁵³See Nesset and Nordrum 2019: 91 for corpus evidence that most of these nouns display the unusual stress ad-paucally in around 80% of text occurrences, with the exception of *šag* 'step', which has this non-SG.GEN stress in around 40% of occurrences.

like PL.GEN. Moreover, adjectives modifying the supposedly SG.GEN noun have PL.GEN forms, as in (50), in an apparent violation of concord.

- (50) dva krasnyx stula (RUS)
 two.M.NOM/ACC red.PL.GEN(?) chair.M.SG.GEN(?)
 ‘2 red chairs’

While the examples above involve masculine nouns, neuter nouns behave similarly, i.e., even though they seem to be SG.GEN ad-paucally, they are modified by apparently PL.GEN adjectives in contemporary Russian. By contrast, feminine nouns occur ad-paucally most often with plural adjectives syncretic between NOM and ACC (but not GEN), e.g.:

- (51) dve krasnye krovati (RUS)
 two.F.NOM/ACC red.PL.NOM/ACC(?) bed.F.SG.GEN(?)=PL.NOM/ACC
 ‘2 red beds’

Various sources give different generalizations about the co-occurrence of adjectives and nouns in ad-paucal positions. This is probably because, in such positions, both PL.NOM/ACC and PL.GEN adjectives could occur with apparently SG.GEN nouns bearing different genders to various degrees at various stages of Modern Russian. However, the detailed diachronic corpus investigation of this matter in Nessel 2020 demonstrates that the percentage of PL.GEN adjectival forms modifying masculine and neuter ad-paucal nouns grew steadily from around 15% in mid-XIXth century to 99% in early XXist century. In the case of feminine nouns, the trend was reverse: the percentage of PL.GEN forms diminished from around 80% at the end of XIXth century to 13% in early XXist century. Hence, the contemporary generalization is that 1) only PL.GEN adjectives may occur with masculine or neuter nouns, while 2) both PL.NOM/ACC and PL.GEN adjectives may occur with feminine nouns, but with a strong and growing preference for PL.NOM/ACC. See Nessel 2020 for additional analyses.

Stress also complicates the SG.GEN analysis of feminine nouns. While many feminine nouns have syncretic SG.GEN and PL.NOM/ACC forms (e.g., *krovati* in (51)), some distinguish these two possibilities by stress pattern, e.g., *storóný* ‘side.SG.GEN’ vs. *stóroný* ‘side.PL.NOM/ACC’. In such cases, usually the apparently SG.GEN form occurs ad-paucally, regardless of the form of the accompanying adjective, but the PL.NOM/ACC stress pattern is also occasionally observed, especially in the case of *stóroný* ‘side.PL.NOM/ACC’ (Nessel and Nordrum 2019). Finally, as frequently noted (Franks 1994, 1995, Pesetsky 2013, Pereltsvaig 2010, Nessel and Nordrum 2019), also de-adjectival feminine nouns occur ad-paucally in what looks like plural rather than the expected singular, e.g.:

- (52) tri {masterskie / masterskix} / Ivanovy / Ivanovyx (RUS)
 three.F.NOM/ACC workshops.F.PL.{NOM/ACC / GEN}(?) Ivanova.F.PL.{NOM/ACC / GEN}(?)
 ‘3 {workshops / Ivanovas}’ (where *Ivanova* is an inflectionally adjectival surname)

For all these reasons, contemporary analyses of Russian paucal constructions assuming that the noun is indeed SG.GEN are very rare (but see Nessel and Nordrum 2019). Instead, most analyses posit that ad-paucal nominals have a feature which is not observed elsewhere in Russian. The question is whether this unique feature is a special grammatical case, or a special grammatical number (or perhaps both; cf. Pereltsvaig 2010).

The “paucal case” hypothesis has a long tradition in Russian linguistics (see, e.g., Zaliznjak 1967: 46–48), but it is also found in some more contemporary generative accounts (e.g., Franks 1995, Yadoff 1999, Rappaport 2002, Ionin and Matushansky 2018). As shown in §5.2.1 below, adopting this hypothesis makes an analysis of case patterns in Russian paucal constructions trivial. However, it does not solve the conceptual problem that – despite the very robust generalization that Russian does not distinguish genders in the plural – the form of plural “paucal adjectives” would depend on gender: they would be syncretic with genitive in masculine and neuter, and preferentially syncretic with the nominative in feminine.

The “paucal number” hypothesis seems to be more frequently assumed in western linguistics (e.g., Rakhlin 2003, Bailyn and Nevins 2008, Asinari 2019, Breu 2020; cf. Corbett 1993: 14–16).⁵⁴ It is more attractive cross-linguistically, given that paucal number – but not paucal case – is witnessed in other languages, and it is perhaps more compatible with diachrony, given that the development of the paucal construction was triggered by the loss of dual number in Old Russian (Nessel 2020), but it faces the conceptual problem that, unlike in languages with true paucal number, the apparently paucal forms of Russian nouns cannot be used alone – in the absence of paucal numerals – to express semantic paucal number. We will see in §5.2.2 that accommodating this hypothesis calls for interesting extensions to the account of the previous sections.

Works advocating either “paucal case” or “paucal number” are sometimes less explicit about the value of the other category in paucal constructions. For example, if the case of *stula* ‘chair’ in *dva stula* ‘2 chairs’ is paucal, what is its number – singular or plural? Similarly, if *stula* has paucal number, what is its case in NOM/ACC positions? Below, we will

⁵⁴See also Pesetsky 2013 for a different analysis in terms of exceptional number.

assume values of the other categories that make the resulting analyses maximally similar to the standard analysis of higher numerals (but we will also mention alternatives). This means that, on the “paucal case” hypothesis, the number of *stula* is plural in NOM/ACC, as it is in other case positions and in the case of higher numeral constructions; see the initial columns of the table in (53).⁵⁵ Similarly, on the “paucal number” hypothesis, the case of *stula* in NOM/ACC positions is taken to be GEN, as in the case of higher numerals; see the final columns in (53).⁵⁶

(53) Values of number and case on the two paucal hypotheses:⁵⁷

position	“paucal case”		“paucal number”	
	‘two’	‘chairs’	‘two’	‘chairs’
NOM	<i>dva</i> . NOM	<i>stula</i> . PL.PCL	<i>dva</i> . NOM	<i>stula</i> . PC.GEN
ACC	<i>dva</i> . ACC	<i>stula</i> . PL.PCL	<i>dva</i> . ACC	<i>stula</i> . PC.GEN
GEN	<i>dvux</i> . GEN	<i>stul’ev</i> . PL.GEN	<i>dvux</i> . GEN	<i>stul’ev</i> . PL.GEN
DAT	<i>dvum</i> . DAT	<i>stuljam</i> . PL.DAT	<i>dvum</i> . DAT	<i>stuljam</i> . PL.DAT
LOC	<i>dvux</i> . LOC	<i>stuljax</i> . PL.LOC	<i>dvux</i> . LOC	<i>stuljax</i> . PL.LOC
INS	<i>dvumja</i> . INS	<i>stuljami</i> . PL.INS	<i>dvumja</i> . INS	<i>stuljami</i> . PL.INS

5.2.1 Paucal Case⁵⁸

A formalization of the “paucal case” hypothesis in terms of the analyses of previous sections is immediate. The only difference between higher numerals (or the Q head associated with them) and the paucal numerals (or the relevant Q) is the case value with which they flag their complements: GEN is the case associated with higher numerals, PCL is associated with paucal numerals:

(54) Russian higher numerals: [INT:ϕ, FLAG:GEN]

(55) Russian paucal numerals: [INT:ϕ, FLAG:PCL] (“paucal case” hypothesis)

On this hypothesis, and assuming case stacking and the *Expone Most Marked* rule as in Polish, there are two case values stacked on ad-paucal nouns: PCL and the case specific to a given syntactic position. When that second case is NOM or ACC, the more marked PCL is expounded, but when the second case is an oblique (GEN, DAT, LOC, INS), that more marked oblique is expounded. Essentially this analysis is proposed in Ionin and Matushansky 2018: §§6.2–6.3.

This reasoning crucially depends on PCL being situated between ACC and GEN in an extended case hierarchy, as shown in (57):

(56) NOM < ACC < GEN < DAT < INS < . . . (Caha 2009; repeated from (30))

(57) NOM < ACC < **PCL** < GEN < DAT < INS < . . . (extended case hierarchy)

Fortunately for the “paucal case” hypothesis, case syncretism provides arguments for exactly this position of PCL.

Recall that, according to Caha 2009, only neighboring cases may display systematic syncretisms. Given that Russian attributive adjectives agree in case with the nouns they modify, not only ad-paucal nouns, but also ad-paucal adjectives must be assumed to bear PCL. For example, according to the “paucal case” hypothesis, the slightly extended examples (50)–(51) have morphosyntactic features as in (58)–(59), with # marking the rarer option.

(58) dva {krasnyx / *krasnye} stula (RUS)
two.M.NOM/ACC red.M(=N).PL.PCL(= {GEN / *NOM/ACC}) chair.M.PL.PCL(=SG.GEN)
‘2 red chairs’

(59) dve {krasnye / #krasnyx} krovati (RUS)
two.F.NOM/ACC red.F.PL.PCL(= {NOM/ACC / #GEN}) bed.F.PL.PCL(=SG.GEN=PL.NOM/ACC)
‘2 red beds’

As illustrated in (58), a PL.PCL form modifying a masculine (or neuter) noun must be syncretic with PL.GEN, while as illustrated in (59), a PL.PCL form modifying a feminine noun is normally syncretic with PL.NOM/ACC, but with forms syncretic with PL.GEN also encountered. Hence, PCL must be a direct neighbor of both GEN and NOM/ACC, as in (57).

While the “paucal case” hypothesis is immediately compatible with the approach to concord and NCCs advocated above, it is far from clear that it is true. One conceptual problem is that, on this hypothesis, the robust generalization about Russian that there are no gender distinctions in plural is lost. This is because – as shown in (58)–(59) – different

⁵⁵This is the explicit position of Ionin and Matushansky 2018: §6.3.6.

⁵⁶This is the explicit position of Asinari 2019, but see also references therein for alternatives and §5.3 for discussion.

⁵⁷In order to keep with the convention of abbreviating case values with 3 letters and number values with 2 letters, and to decrease ambiguity, we abbreviate paucal case to PCL and paucal number to PC.

⁵⁸The Russian term *sčëtnyj padež*, lit. ‘count case’, has been variously translated as “numeral” case (e.g., Nessel and Nordrum 2019) or as “adnumeral”, “countative” or “calculative” case (various web pages). However, in most of the English-language generative literature it is simply called “paucal” case (e.g., Franks 1994, 1995, Rappaport 2002, Ionin and Matushansky 2018; cf. “adpaucal case” in Yadroff 1999: 146), so we also use this term here.

forms of PL.PCL adjectives occur with masculine/neuter nouns (only those that are syncretic with PL.GEN) and with feminine nouns (mainly those that are syncretic with PL.NOM/ACC). The “paucal number” hypothesis is free from this problem, as demonstrated in the next subsection. Another conceptual argument against “paucal case” is that, unlike the cross-linguistically widely attested paucal number, “paucal case” does not seem to occur anywhere beyond East Slavic. Moreover, the “paucal number” hypothesis meshes better with the diachronic developments of Russian paucal constructions, which started with the gradual loss of the dual number during the Old Russian period, as convincingly argued in Nessel 2020; a reanalysis of the last vestiges of dual number as paucal number is less dramatic than a reanalysis of a vanishing number as a totally new case, not observed elsewhere in Russian or beyond.

Finally,⁵⁹ it is a robust cross-linguistic generalization that, unlike pronouns, lexical nouns do not exhibit case-based root suppletion and show only very limited stem allomorphy (limited to NOM and perhaps ACC, if it is syncretic with NOM); see, e.g., Moskal 2015, McFadden 2018, and Smith *et al.* 2019. By contrast, on the “paucal case” hypothesis, case-based root suppletion would be observed which would distinguish PCL from all other cases in the plural:

- (60) a. {rebēnok / rebēnka / – / rebēnku / rebēnke / rebēnkom} (RUS)
 child.SG.{NOM / ACC=GEN / PCL / DAT / LOC / INS}
 b. {deti / detej / **rebēnka** / detjam / detjax / det'mi} (RUS)
 children.PL.{NOM / ACC=GEN / **PCL** / DAT / LOC / INS}
 c. {dva rebēnka / pjat' detej} (RUS)
 two.NOM/ACC children.PL.PCL five.NOM/ACC children.PL.GEN
 ‘{two children / five children}’

As shown in (60a–b), the singular root for ‘child’ in Russian is *rebēnk*, while the plural root is normally *det*. The single exception is the hypothetically PL.PCL form *rebēnka*. This is not a matter of a special ad-numeral form, as shown in (60c): the ad-numeral plural genitive is the regular *detej*.⁶⁰ Hence, the only explanation for this suppletion on the “paucal case” analysis is that it is case-driven, contrary to the generalization mentioned above. Additionally, this analysis would result in a clear violation of the robust *ABA generalization (on which see, e.g., Bobaljik 2012 and §6.4 below), with the root in PL.PCL (the B in *ABA) different from the roots in surrounding cases in plural (the As in *ABA).⁶¹ By contrast, on the “paucal number” analysis below, this root suppletion is number-driven, with SG and PC forms based on *rebēnk*, and PL forms based on *det*.⁶²

While none of these problems alone is necessarily fatal for the “paucal case” hypothesis, together they justify an attempt at substantiating the alternative “paucal number” hypothesis.

5.2.2 Paucal Number

On the analysis proposed in this subsection, the exponed features of the two running examples (50)–(51) are as indicated in (61)–(62).

- (61) dva {krasnyx / *krasnye} stula (RUS)
 two.M.NOM/ACC red.PL.{GEN / *NOM/ACC} chair.M.PC.GEN(=SG.GEN)
 ‘2 red chairs’
 (62) dve {krasnye / #krasnyx} krovati (RUS)
 two.F.NOM/ACC red.PL.{NOM/ACC / #GEN} bed.F.PC.GEN(=SG.GEN=PL.NOM/ACC)
 ‘2 red beds’

⁵⁹Apart from these problems, the “paucal case” analysis also does not predict the following exponence contrast between hypothetically paucal nouns and adjectives: while there exist hypothetically PCL nouns that are *not* syncretic with SG.GEN, which justifies an analysis in terms of some special paucal feature, hypothetically PCL attributive adjectives are without exception syncretic with PL.GEN or – in the case of feminine nouns – PL.NOM/ACC. On the “paucal number” analysis below, the adjectives are what they seem and such an exceptionless syncretism does not need to be invoked.

⁶⁰Such a special ad-numeral plural form is witnessed in the case of *čelovek* ‘person.SG’, which in the plural is normally based on the root *ljud* (e.g., *ljudi* ‘people.PL.NOM’), but in ad-numeral contexts, in NOM, ACC, and GEN positions, it reverts back to the singular root *čelovek* (e.g., *dva čeloveka* ‘two people’, *pjat' čelovek* ‘five people’). On the current analysis, this suppletion would be conditioned by the feature NUM(eral), rather than by case (with GEN contexts requiring some additional explanation). Similarly, *god* ‘year’, has ad-numeral forms showing the root suppletion *let* (normally a PL.GEN form of *leto* ‘summer’), but only in plural, not in paucal (e.g., *odin god* ‘one year.SG.NOM’, *dva goda* ‘two years.PC.GEN’, but *pjat's let* ‘five years.PL.GEN’). On the current analysis, this form would expose the feature bundle containing [NUM, PL], i.e., it would be conditioned on category (resulting from flagging by the numeral) and number – not on case. Compare Ionin and Matushansky 2018: §6.6.2, from which the suppletion data discussed here is drawn, as well as Pancheva 2018, Franks 2018, and Ionin and Matushansky 2018: §6.6.3 on ad-numeral forms in Bulgarian (a language with overt case limited to pronouns).

⁶¹*ABA is violated here not only in terms of root suppletion, but also in terms of syncretism, given that, in (60b), (ACC.PL = GEN.PL) ≠ PCL.PL, and that the ACC < PCL < GEN part of the case hierarchy in (57) is crucial for the “paucal case” explanation of case patterns in Russian NNCs.

⁶²See fn. 68 on the possibility that, in Russian, PL is more marked than SG and PC. Moreover, given that such feature hierarchies may be partially rather than linearly ordered (see, e.g., Zoppi 2023 on person and §6.4 below and references there on case), another possibility is that both PC and PL are more marked than SG, with no order between PC and PL. On either analysis of Russian number, our proposal that suppletion may target PL to the exception of SG and PC does not violate the *ABA generalization.

Note that, on this analysis, the forms of adjectives are what they seem: plural genitive (rather than paucal genitive or plural paucal) in (61) and PL.NOM/ACC (rather than PC.NOM/ACC or PL.PCL) in (62).

The main idea of the analysis of Russian paucals offered below is this:

- paucal numerals – and only they – enter the derivation specified for the paucal number,⁶³
- nouns in quantity contexts enter the derivation specified for the plural;
- as a result of the numeral–noun Agree, each ϕ -feature bundle is copied to the other item;
- this way nouns end up with the paucal number, but only in ad-paucal positions.

This idea immediately accounts for the conceptual problem with the “paucal number” hypothesis mentioned above: if Russian has paucal number, why cannot paucal forms of nouns be used alone to express paucal quantity (Corbett 1993, 2000: 22–23, fn. 18)? For example, why *vladelec stula* ‘owner.NOM chair.GEN.PC=SG’ means only ‘owner of a chair’ and cannot be used to express ‘owner of a couple of chairs’? The answer is a lexical idiosyncrasy grounded in diachrony: the only lexical items in contemporary Russian that bear the paucal number are the paucal numerals, born on the ashes of dual (Nesset 2020). Nouns may receive paucal number only when they are probed by such paucal numerals, so only then can they express paucal quantity. Hence, the analysis offered here is free from the main conceptual problem with the “paucal number” approach.

In order to substantiate this analysis, let us first recall that, as a separate grammatical category, Slavic numerals stem from adjectives (paucals) and nouns (higher numerals), and share certain features with both these more established categories. Words expressing smaller numbers display more adjectival behavior and those expressing higher numbers have more features common with nouns (Corbett 1978), to the extent that *odin* ‘one’ is a pure adjective (in particular, it inflects for number, unlike true numerals, e.g., *odni štany* ‘a pair of trousers’, lit. ‘one.PL trousers.PL’) and *million* ‘million’ is sometimes claimed to be a pure noun (in particular, it has a specific – masculine – gender and it takes a genitive noun even in oblique cases, both unlike true numerals).⁶⁴

Given that Slavic numerals stem from adjectives and nouns, we posit that their featural makeup is somewhere between that of adjectives and that of nouns. A conspicuous difference between these two categories in north Slavic is that adjectives do not have any lexically-specified inherent ϕ features, while nouns are lexically inserted with an idiosyncratic gender and sometimes idiosyncratic number, the latter conspicuous in *plurale tantum* and *singulare tantum* nouns (e.g., *štany* ‘trousers.PL’ and *kto* ‘who.SG’). We claim that, by comparison, north Slavic numerals are an intermediate class: they are lexically specified for number, but not for gender. Additionally, most numerals are more like nouns in combining with GEN complements, while some paucals are more like adjectives in the lack of GEN-flagging. However, even when numerals GEN-flag their NP complements, this genitive is overridden by more marked oblique cases, unlike in the case of nouns, whose GEN complements are KPs impenetrable to further case flagging. This is summarized in (63).

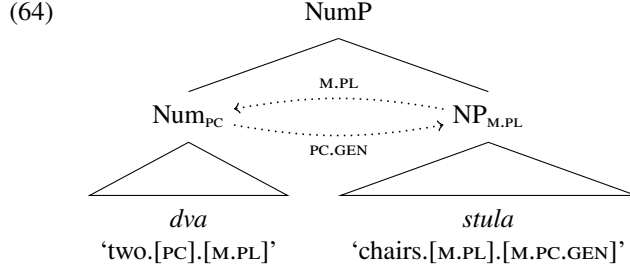
(63)	Adjectives (incl. <i>odin</i> ‘1’)	N u m e r a l s			N o u n s (incl. <i>million</i> ‘10 ⁶ ’)
		P a u c a l s	High		
		adjectivish	nounish		
lexical number	–	+	+	+	(+)
GEN complements. . .	–	–	+	+	+
. . . also in oblique	–	–	–	–	+
lexical gender	–	–	–	–	+

What is special about Russian paucals is that their lexically specified number is, well, paucal (diachronically stemming from the old dual). Specifically, we claim that paucal numerals are the only lexical items specified for PC in contemporary Russian; in particular, higher numerals are specified as plural (similarly to all numerals in Polish) and nouns are lexically inserted as singular or plural.

Given that Russian countable nouns are typically plural in quantity environments, regardless of the number of the selector (e.g., *para stul’ev* ‘a pair of chairs’, lit. ‘pair.F.SG.NOM chairs.M.PL.GEN’), the first step in the construction of a simple paucal phrase, *dva stula* ‘two chairs’, is as in (64).

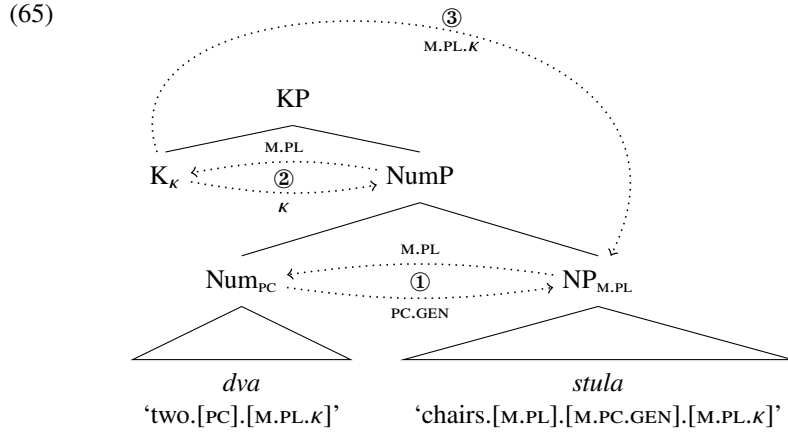
⁶³For simplicity, we assume throughout this paper that items in the lexical array may be specified for features such as gender and number. Nothing seems to hinge on this assumption and the analysis offered here is compatible with views on which all grammatical features are introduced by functional heads. For example, the statement that in Russian only and all paucal numerals are lexically specified as bearing the paucal number may be translated into the property of the Number/# functional head that it may bear the paucal feature only in the context of roots of paucal numerals.

⁶⁴In fact, Russian *million* may be ambiguous between a noun and a numeral (Ionin and Matushansky 2018: §6.2.6), similarly to Polish *milion* ‘million’ (Przepiórkowski 1999: §5.3.2.3).



As explained above, the numeral enters the derivation specified as paucal, [PC], and the noun as plural, here [M.PL]. There are two results of Agree between them: copying of the noun’s ϕ features to the numeral, and flagging the numeral with GEN, but also – as in the system of Clem and Deal 2024 – with the numeral’s ϕ features.⁶⁵ Hence, each item has two feature bundles, as indicated on the terminals in (64).⁶⁶ These feature bundles form a stack; this is a natural generalization not only of the analysis of NNCs in §4 in terms of case stacks, but also of the analysis of Grishin 2024, on which ϕ features agglomerated as a result of multiple Agree form a stack. While all stack elements are visible by morphological exponence rules, we assume with previous literature that syntactic processes only see the top of the stack.⁶⁷ For example, once the numeral agrees with the nominal as in (64), the feature bundle on Num (and, hence, also on NumP) active in ensuing syntactic processes will be [M.PL], which is on top of the stack, and not the original [PC] at the bottom of the stack.

The result of the next step, merge with K, is shown in (65):



After the initial numeral–noun Agree step discussed above, the ϕ features on the top of Num(P)’s stack are [M.PL], so it is this bundle that gets copied to K from NumP in the second step, when the NumP is flagged with κ . Hence, subsequent flagging involves the [M.PL. κ] bundle, resulting in the presence of this bundle on the top of N(P)’s stack. If κ is NOM or ACC, the [M.PC.GEN] bundle is more marked, so the paucal number is expounded on the genitive noun. By contrast, when κ is DAT, LOC, or INS, the outermost bundle involving κ is the one that is most marked, so it is expounded, with the appropriate case and ordinary plural number (originally introduced by the noun itself!), as shown in the table in (53) and, for DAT, in (66) below.

- (66) dvum {stuljam / *stula} (RUS)
 two.DAT chair.M. {PL.DAT / *PC.GEN(=SG.GEN)}
 ‘2 red chairs’

Interestingly, when κ is GEN and, hence, the two cased feature bundles on the noun are the paucal [M.PC.GEN] and, on the top of the stack, the plural [M.PL.GEN], it is the outermost bundle that is expounded – the noun has the plural genitive form, not the paucal genitive form:

- (67) dvux {stul’ev / *stula} (RUS)
 two.GEN chair.M. {PL.GEN / *PC.GEN(=SG.GEN)}
 ‘2 red chairs’

⁶⁵We ignore the possibility of category flagging, as it does not seem to play any role here, but it may be crucial to explain some ad-numeral forms, e.g., the suppletive *let* lit. ‘summers.PL.GEN’ used to expound the ad-numeral ‘year.PL.GEN’ instead of the regular *godov* ‘year.PL.GEN’.

⁶⁶The presence of M in [M.PC.GEN] on the right terminal is explained in Appendix A.2. As before, subscripts on pre-terminals (here, on Num and N) indicate the inherent (lexical) features of the relevant items (but cf. fn. 63).

⁶⁷See, e.g., Bruening and Al Khalaf 2020 and Bruening 2023 for analyses employing stacks and assuming that certain processes only see the top of a stack (but also see Patejuk and Przepiórkowski 2023 and Przepiórkowski and Patejuk 2025 for a critique of these particular analyses).

There are various ways to ensure this result.⁶⁸ The analysis that we adopt here for concreteness is that markedness is determined solely by case, and when there is a draw, the outermost of the maximally marked bundles is expone. That is, we replace the *Expone Most Marked* rule in (28) with the *Expone Outer Most Marked* rule in (68).⁶⁹

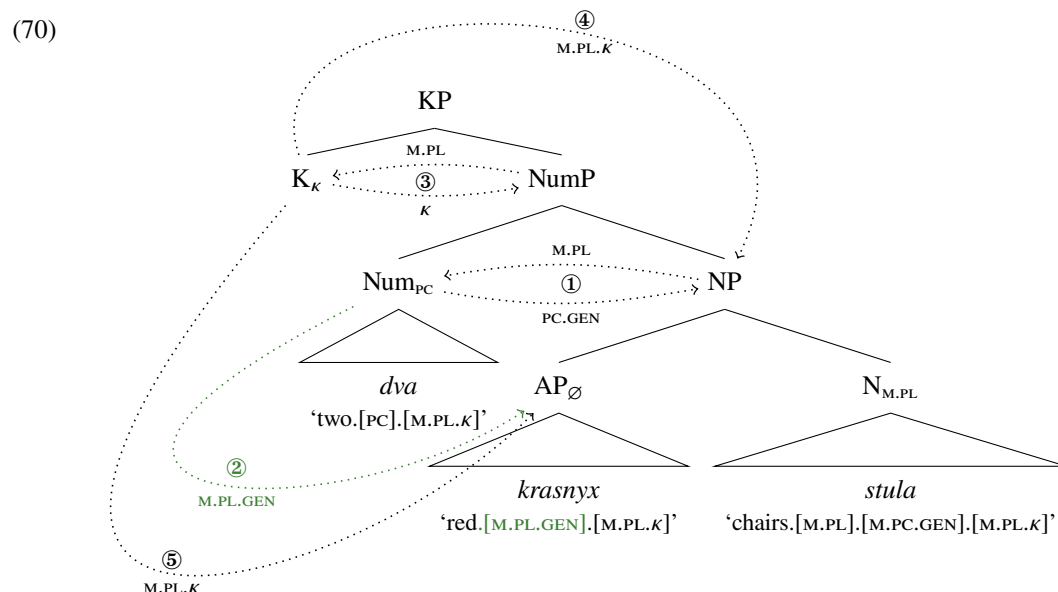
(68) *Expone Outer Most Marked*: Expone only the outermost of maximally marked feature bundles.

Note that, when there is a single bundle that is most marked, (68) has the same effect as the *Expone Most Marked* rule in (28) assumed so far.

The analysis above does not depend on the gender of the noun, but such a dependence is needed when adjectival modifiers are considered. It is at this point that we need to be more specific about the nature of the probe on Num.

Consider the tree in (70) for the paucal phrase in (61), partially repeated below as (69), which differs from the previous tree in (65) in the presence of an adjectival modifier. Steps marked ①, ③, and ④ are exactly the three steps considered above. Additionally, given that the probe on K is insatiable, it also flags the AP with [M.PL.K], as discussed in §3 (see step ⑤). The new question, one that we have ignored so far, is whether the probe on Num should also be insatiable, as assumed in (70) (see step ② and the related bits in green), or not.

(69) dva krasnyx stula (RUS)
two.M.NOM/ACC red.PL.GEN chair.M.PC.GEN(=SG.GEN)
'2 red chairs'



If the probe on Num is insatiable, it does not stop after Agree with the NP, but it also flags the AP with [M.PL.GEN] (as, after step ①, M.PL is on top of Num(P)'s stack), so this feature bundle is added to the A(P)'s stack in step ②, before [M.PL.K] is added in step ⑤. This predicts that the AP is expone as PL.GEN in NOM/ACC positions, as [M.PL.GEN] is more marked than [M.PL.K] then, and as [M.PL.K] in other positions. This is the right prediction for all masculine and neuter nouns (as illustrated in the table in (53)), and it is still a possibility for feminine nouns.

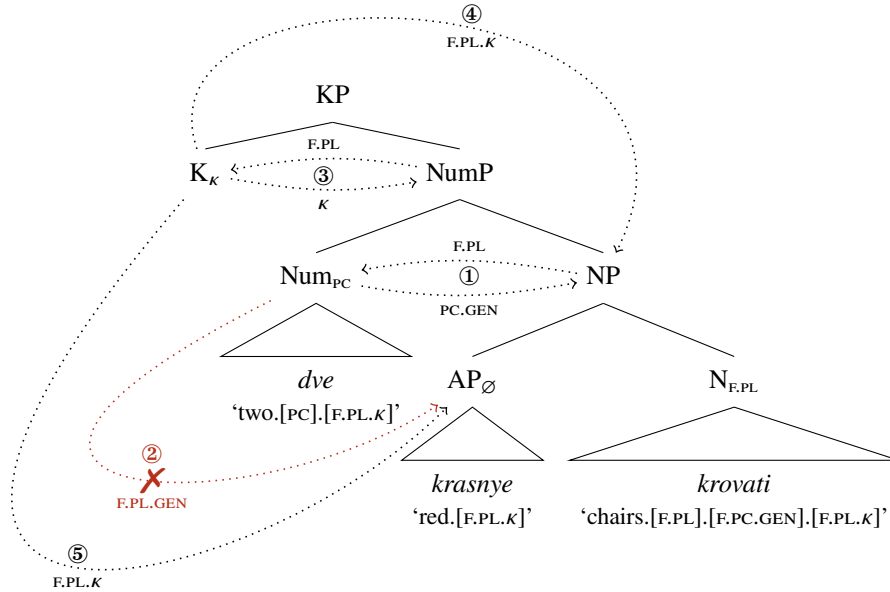
But what about the more common situation in feminine paucal constructions, where APs modifying the F noun are in NOM/ACC? It turns out that this possibility is readily accommodated by the system proposed here, on the assumption that in such cases the probe is immediately satiable. Then the step ② does not take place, and the result is as in the tree in (72) for the paucal phrase in (62), partially repeated in (71).

(71) dve krasnye krovati (RUS)
two.F.NOM/ACC red.PL.NOM/ACC bed.F.PC.NOM(=SG.GEN=PL.NOM/ACC)
'2 red beds'

⁶⁸One would be to assume a markedness metric on which, at least in Russian, plural is more marked than paucal and, hence, the feature bundle [M.PL.GEN] is more marked than [M.PC.GEN]. This seems to go against the commonly assumed hierarchy SG < PL < DU/PC (see, e.g., Nevins 2011 and references therein), but – as noted in Smith *et al.* 2019 – there exist languages where suppletion facts suggest that dual is *less* marked than plural, one of them being Slovenian. So, given that both Slovenian dual and Russian paucal stem from Proto-Slavic dual, the possibility that Russian plural is more marked than paucal should not be rejected without investigation.

⁶⁹The name of this rule is the result of *of*-deletion and haplogy of *most* in *Expone Outermost of Most Marked*.

(72)



Note that the only feature bundle on the A(P) is now [F.PL.K], so such modifiers are NOM/ACC in a NOM/ACC environment, as illustrated in (71), even though the feminine noun is expounded as [F.PC.GEN] – usually the same as SG.GEN rather than PL.NOM, where suprasegmental features make this distinction visible.

The following summarizes the properties of probes on Num (or the associated Q) in Slavic numeral constructions dealt with above ((73) and (75) replace the earlier (46)–(47)):

- (73) Russian/Polish higher numerals: [INT: ϕ , SAT:–, FLAG:GEN], SEL:N
 (74) Russian paucal numerals: [INT: ϕ , SAT:–, FLAG:GEN], SEL:N
 [INT: ϕ , SAT: ϕ , FLAG:GEN], SEL:N.F
 (75) Polish paucal numerals: [INT: ϕ , SAT:–, FLAG:GEN], SEL:N.m1
 [INT: ϕ , SAT:–, FLAG: —], SEL:N

The most widespread Num probe is [INT:φ, SAT:–, FLAG:GEN]: in Russian, it is present on both paucal and higher numerals and it does not constrain selectional restrictions, while in Polish it is present on higher numerals and, with restriction to m1, on paucals. While we justified its insatiability on the basis of Russian paucals, it must also be insatiable on higher numerals (and Polish paucals combining with m1 nouns), given that in NOM/ACC positions not only the nouns but also their adjectival modifiers must be GEN, e.g.:

- (76) Przeczytał pięć {dobrych / *dobre} książek. (POL)
 read_{ACC.PST.M.3SG} five.F.ACC good.F.PL.{GEN / *ACC} book.F.PL.GEN
 'He read 5 good books.'

Additionally, there are two Num probes that differ from this one minimally. The additional Russian probe, present on paucals selecting for N.F, differs minimally in being immediately satiable, as discussed above. The additional Polish probe, present on all paucals, differs in the absence of GEN-flagging.⁷⁰ All these probes conform to recent approaches to Agree exemplified by Clem and Deal 2024, and they make a crucial use of the satiability aspect of these approaches.

5.3 Interim Summary and Comparisons

The analysis of Polish and, especially, Russian paucal numerals offered in this section builds on the general account of concord as insatiable Agree in §3 and on the particular application of this account to Slavic numerals in §4, which makes use of case stacking. This idea was extended in subsection §5.2.2 to feature bundle stacking, in order to account for correlations between case and number in Russian paucals: according to this analysis, the noun bears the paucal number when the whole paucal phrase is in a *NOM/ACC* position, but the plural number in other case positions. The analysis also formally models the forms of modifying adjectives in *NOM/ACC* positions (genitive plural or, only in feminine, nominative/accusative plural) without stipulating any systematic syncretisms. The analysis is also free from known

⁷⁰It is SAT:– in (75), but – given the lack of GEN-flagging – SAT:ϕ would also work for this probe, and it is not clear how to choose between these two options. (If NumP could be a maximal nominal projection in some environment, to the exclusion of DP or KP layers, then this insatiability would be needed to flag adjectival modifiers with an appropriate number and gender, but it is not clear to us whether NumP can ever be such a maximal extended projection.)

conceptual problems with either paucal case or paucal number, which we believe makes it attractive in the landscape of analyses of Slavic paucals proposed so far.

While concentrating on the “paucal number” hypothesis, we noted that also the “paucal case” approach is immediately formalizable within the concord framework advocated in this paper. Nevertheless, we pointed out various problems with “paucal case”, which gave impetus for developing the “paucal number” analysis.

According to the “paucal number” analysis proposed above, the noun is in the genitive in NOM/ACC positions, just as in constructions headed by higher numerals. We share this perspective with Asinari 2019 (as well as with Pesetsky 2013, see below), but there are also two “paucal number” analyses which assume that the noun is actually NOM/ACC, not GEN, in such positions: Rakhlin 2003: §5 and Bailyn and Nevins 2008: §8.6.⁷¹ However, the arguments given for this position are not convincing. In the case of Bailyn and Nevins 2008, the main argument is from the possibility of agreement of the paucal construction with the verb, which implies that the paucal construction is in the nominative. This is not controversial. However, Bailyn and Nevins (2008) further claim that paucal numerals cannot be the NOM-bearing heads, as some such numerals inflect for gender, which makes them adjectival and, as such, non-heads. Hence – the reasoning goes – it must be the noun that is in the nominative. However, this argument does not follow: pure adjectives are heads in more complex adjectival constructions, e.g. (Babby 1975: 34–35):

- (77) {dolžna / dolžen} Maše bol’šuju summu (RUS)
 due.F/M Masha.DAT large.ACC sum.ACC
 ‘owing.A.F/M Masha a large sum’

Additionally, the analysis in Rakhlin 2003 is based on an incorrect generalization, namely, that ad-paucal nouns in ACC positions are modified by ACC adjectives, while in fact all masculine and neuter nouns are modified by GEN adjectives, and only in the case of animate nouns, which are the focus of Rakhlin 2003, may this GEN be perhaps interpreted as (animate) ACC. In short, we are not aware of any convincing arguments against the analysis of paucal constructions in §5.2.2, on which the noun bears the genitive in NOM/ACC positions, just as in other numeral constructions.

Let us finally note that, while – as mentioned above – we share with Pesetsky 2013 the analysis of nouns in NOM/ACC paucal constructions as truly genitive, we do not agree with another crucial aspect of that account. For Pesetsky (2013), such nouns do not bear paucal number, but are actually numberless; what is usually taken to be a paucal *numeral* is actually a grammatical *number morpheme*. This, however, not only reintroduces the dual number (given the presence of the number morpheme *dva* ‘two’), generally considered long lost in Russian, but also introduces the cross-linguistically rare trial (*tri* ‘three’), the highly questionable (Corbett 2000: §2.2.5) quadral (*četyre* ‘four’), as well as hitherto unknown grammatical numbers corresponding to *poltora* ‘one and a half’, and also to *pol-* ‘half’ and *četvert’* ‘quarter’.⁷² Instead of introducing all these number values into Russian grammar, some of which have not been attested anywhere else so far, we posit that Russian simply has some vestiges of paucal, which is cross-linguistically well-attested (Corbett 2000: §2.2.4).

6 Heterogeneous Patterns in Finnic

Norris (2018) claims that Estonian pseudopartitives are a challenge for analyses of heterogeneous patterns based on case stacking. In this section, we first present the relevant Estonian data (in §6.1) and the challenge for case-stacking accounts (in §6.2). We show (in §6.3) that while the relevant data are problematic for an account based on the *Exponere Outermost* rule, it is not for our version of exponence rules, based on markedness. We motivate (in §6.4) our modification of Caha’s (2009) case hierarchy needed to account for Estonian pseudopartitives and we finish (in §6.6) with some cross-linguistic considerations, including a brief comparison to another Finnic language, Finnish.

6.1 Estonian Pseudopartitives

Pseudopartitive constructions contain two nouns, the first of which (N1) indicates the quantity of the entity or entities denoted by the second (N2), as in English *piece of bread* or *bag of potatoes*. In Estonian, the two nouns bear the same case in most syntactic positions, with the exception of nominative and accusative, as illustrated with the Estonian for ‘piece of bread’ in (78).

Two aspects of this table do not follow tradition. First, Norris (2014, 2018) convincingly argues that four of traditional Estonian cases – terminative, essive, abessive, and comitative (shown at the bottom of the table) – are actually clitic postpositions combining with a nominal phrase in the genitive. That is why these “cases” are only marked on the last word of the phrase, N2, rather than on both N1 and N2.⁷³ Second, the Estonian accusative is not usually recognized in traditional grammars, as it is not autonomous: it is always syncretic either with genitive (in the singular, as in (78)) or with nominative (in the plural). In this respect, Estonian is very similar to Finnish, which however has some autonomous (i.e.,

⁷¹This suggestion appears also in Corbett 1993: 15.

⁷²See Ionin and Matushansky 2018: §§6.3.7 and 6.4.3 on the paucal status of these two words.

⁷³See Weisser 2020 for an analogous claim based on Estonian coordinate structures.

non-syncretic) accusative forms of pronouns. In Estonian, there is even less evidence for accusative as a separate case, but a convincing argument may be found in Norris 2014, 2018.⁷⁴

(78)	position	‘piece.sg’	‘bread.sg’
	NOM	tükk. NOM	leiba. PAR
	ACC	tüki. ACC	leiba. PAR
	PAR	tükki. PAR	leiba. PAR
	GEN	tüki. GEN	leiva. GEN
	ILL	tüki-sse. ILL	leiva-sse. ILL
	INE	tüki-s. INE	leiva-s. INE
	ELA	tüki-st. ELA	leiva-st. ELA
	ALL	tüki-le. ALL	leiva-le. ALL
	ADE	tüki-l. ADE	leiva-l. ADE
	ABL	tüki-lt. ABL	leiva-lt. ABL
	TRL	tüki-ks. TRL	leiva-ks. TRL
	TER	tüki. GEN	leiva. GEN -ni. TER
	ESS	tüki. GEN	leiva. GEN -na. ESS
	ABE	tüki. GEN	leiva. GEN -ta. ABE
	COM	tüki. GEN	leiva. GEN -ga. COM

Note that the heterogeneous vs. homogeneous pattern observed here is very similar to that in Slavic numerals, but with two important differences. First, the non-agreeing case on N2 in the heterogeneous pattern is partitive rather than genitive, as in Slavic.⁷⁵ Second, the form of N1 in the accusative (here, *tüki*) is not the same as in the nominative (here, *tükk*), while Slavic numerals usually have the same **NOM** and **ACC** forms. This second difference is claimed in Norris 2018 to be incompatible with case-stacking approaches.

6.2 Challenge for Case Stacking?

Norris (2018) considers two variants of the case-stacking approach, differing in the exponence rules:

(79) *Pronounce Outermost*: only the outermost case in a case stack is realized.

(80) *Pronounce Outermost Overt*: Pronounce the outermost case with an overt realization.

The rule in (79) is a paraphrase of Pesetsky’s (2013) *One-Suffix Rule*, which we generalized to the *Expone Outermost* rule in (27), repeated below as (81).

(81) *Expone Outermost*: Expone only the outermost feature bundle. (cf. Pesetsky 2013: 11, (7))

A rule like (79) or (81) suffices in Pesetsky 2013 to expone the genitive on the noun within numeral–noun constructions in nominative and accusative positions, because various special mechanisms make sure that **NOM** and **ACC** are never assigned to the noun within the NNC.⁷⁶ As a result, the case stack of the noun in NNC only contains **GEN**, so this case is exponed.

However, this analysis does not extend to Estonian pseudopartitives, as nominative and accusative have different forms (e.g., *tükk* vs. *tüki* in (78)), so it cannot be claimed that both reflect the lack of external case assignment; even if **NOM** is analyzed as lack of case, **ACC** must be analyzed as a specific case. Hence, after the pseudopartitive N1 flags N2 with **PAR**, the whole construction is flagged with **ACC**, so N2 should have two cases on its stack: the innermost **PAR** and the outermost **ACC**. But then **ACC** should be exponed (e.g., as *tüki leiva* ‘piece.**ACC**=**GEN** bread.**ACC**=**GEN**’, which is grammatical in genitive positions but not in true accusative positions), rather than the actually observed **PAR** (e.g., *tüki leiba* ‘piece.**ACC** bread.**PAR**’, grammatical in accusative positions).

Also the version of the exponence rule in (80), cited in Norris 2018 as assumed in Baker and Vinokurova 2010 and Brattico 2008, 2010, 2011, does not suffice here. Even if Estonian nominative is analyzed as a case without an overt realization – in which case (80) correctly predicts that **NOM** loses to **PAR**, even though **PAR** is deeper on the **PAR.NOM** stack – the same cannot be said about the accusative, so the **ACC** on the **PAR.ACC** stack should be exponed in accusative positions, contrary to facts.

On the basis of this reasoning, Norris (2018: 559) concludes that Estonian pseudopartitives “pose challenges to analyses based on so-called case stacking or multiple case assignment to the quantified phrase”. We take up this challenge

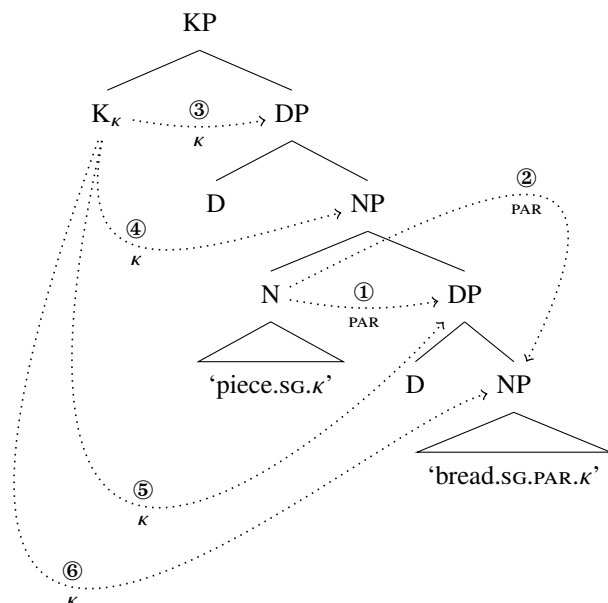
⁷⁴In brief, while N1 has the same form in accusative and genitive positions, N2 is partitive in true (ad-verbal) accusative positions, while it is genitive in true (ad-nominal and ad-prepositional) genitive positions.

⁷⁵In Russian, some nouns have a separate singular form that may be used in some partitive contexts, but the apparently **SG.GEN** form used in paucal constructions is always genitive, not partitive, e.g., *dva {čaja/*čaju}* ‘two teas.**PC.GEN**=**SG**.{**GEN**/***PAR**}’.

⁷⁶These special mechanisms include special properties of Determiners, which normally assign the nominative, but not to the noun within an NNC, and special properties of Verbs, which normally assign the accusative, but not to NNCs.

6.3 Case Stacking Analysis

(82)



However, unlike Norris (2018), we assume that, just as Slavic numerals flag accessible constituents with GEN, Estonian pseudopartitive nouns flag accessible constituents with PAR.⁷⁸ As this happens before K merges, the lower N2 has two cases on its stack, PAR at the bottom and κ at the top (while the higher N1 is only flagged with κ), as indicated in (82).

(83) *Expone Outer Most Marked*: Expone only the outermost of maximally marked feature bundles.

6.4 Estonian Case Hierarchy

(84) NOM < ACC < GEN < DAT < INS < ...

(Caha 2009: 30; repeated from (30) and (56))

(85) NOM < ACC < **GEN** < **PAR** < DAT < INS < ...

(cf. Caha 2009: 115)

⁷⁸That is, both count as semi-lexical categories in the sense of Klockmann 2017.

This hierarchy is only partially compatible with our assumptions. Partitive is more marked than nominative and accusative, accounting for the *PAR* expounded on N2 in *NOM* and *ACC* positions, but it is also supposedly more marked than genitive, predicting *PAR* on N2 also in *GEN* positions, contrary to facts: as shown in table (78), the N2 ‘bread’ is expounded as the *GEN* *leiva* rather than the *PAR* *leiba* in genitive positions.

The remainder of this subsection demonstrates that Estonian morphological facts require a different extension of Caha’s case hierarchy in (84), namely, the extension in (86).

(86) $\text{NOM} < \text{ACC} < \{\text{GEN} \text{ \& } \text{PAR}\} < \text{obliques}$ (Estonian)

On this extension, both *PAR* and *GEN* are more marked than *NOM* and *ACC* and both are less marked than oblique cases, whatever the exact order of markedness within such obliques; these relations are just as in Caha’s (85). However, unlike in (85), *PAR* and *GEN* are not comparable here: neither is more marked than the other.

Note that this hierarchy immediately explains the Estonian heterogeneous vs. homogeneous pattern. In *NOM* and *ACC* positions, the case stacks on N2 are *PAR.NOM* and *PAR.ACC* respectively, so the more marked *PAR* is expounded. In oblique case positions such as illative or inessive, the oblique is more marked than *PAR*, so it is expounded. Crucially, in genitive positions, the case stack on N2 is *PAR.GEN*, and neither of these two cases is more marked than the other, so the *Expone Outer Most Marked* rule repeated below as (87) predicts that the outermost of these – the genitive – is expounded.

(87) *Expone Outer Most Marked*: Expone only the outermost of maximally marked feature bundles.

As illustrated in table (78), this prediction is correct. Hence, if this analysis is on the right track, it provides an independent argument for the *Expone Outer Most Marked* rule – motivated earlier only by the behavior of Russian paucals – as opposed to similar rules found in the literature (*Expone Outermost*, *Expone Most Marked*, etc.).

In order to show that this analysis is on the right track, we need to demonstrate – on the basis of independent morphological facts – that the right case hierarchy for Estonian is the partial order in (86) rather than the total order in (85). Let us first examine the reasoning that led Caha (2009) to the extension in (85). This reasoning is based on Estonian data in tables (88) and (89).⁷⁹

(88) Estonian structural cases, plural

position	‘flag.PL’	‘church.PL’	‘person.PL’
NOM	lipud	kirikud	inimesed
ACC	lipud	kirikud	inimesed
GEN	lippude	kirikute	inimeste
PAR	lippusid	kirikusid	inimesi

(89) Estonian structural cases, singular

position	‘corridor.SG’	‘person.SG’	‘year.SG’	‘church.SG’
NOM	koridor	inimene	aasta	kirik
ACC	koridor-i	inimes-e	aasta	kirik-u
GEN	koridor-i	inimes-e	aasta	kirik-u
PAR	koridor-i	inimes-t	aasta- t	kirik-u- t

Recall first that Estonian *ACC* is systematically syncretic with *NOM* in the plural. This is illustrated in (88). Given that both *PAR.PL* and *GEN.PL* may have different forms than *NOM.PL=ACC.PL*, neither *PAR* nor *GEN* may intervene between *NOM* and *ACC*, so *NOM* must immediately precede *ACC* in this hierarchy: $\text{NOM} <^! \text{ACC}$. Otherwise, if *PAR* or *GEN* intervened between *NOM* and *ACC*, this would lead to a violation of the robust **ABA* generalization.⁸⁰

Similarly, Estonian *ACC* is systematically syncretic with *GEN* in the singular, as illustrated in (89). If *PAR.SG* were also always syncretic with *ACC.SG=GEN.SG*, as in the ‘corridor.SG’ column, then this would tell us nothing about their relative order. However, as shown in the other three columns, *PAR.SG* often differs from *ACC.SG=GEN.SG*. This precludes the order $\text{ACC} < \text{PAR} < \text{GEN}$, as such an order would violate the **ABA* generalization. This in turn means that *GEN* must be adjacent to *ACC*. Given that $\text{NOM} <^! \text{ACC}$, this translates into: $\text{NOM} <^! \text{ACC} <^! \text{GEN}$. So much is uncontroversial.

However, the last two columns of (89) additionally show that *PAR* may morphologically contain *GEN=ACC*, so at least one of the following relations must be true: $\text{ACC} < \text{PAR}$ or $\text{GEN} < \text{PAR}$. But since $\text{ACC} <^! \text{GEN}$, the only possibility seems to

⁷⁹While the content of the tables is drawn from Caha 2009: 114, the order of columns and the graying of cells is slightly different.

⁸⁰In most general terms, “**ABA* generalization” refers to morphological patterns in which, given some arrangement of the relevant forms in a structured sequence, the first and third may share some property ‘A’ only if the middle member shares that property as well” (Bobaljik and Sauerland 2018: 1–2). A common application of **ABA* involves root suppletion and it is best illustrated with an example from degree morphology. Given the hierarchy of degree values: positive < comparative < superlative, *ABB* suppletions such as *good – better – best* are expected, but an *ABA* pattern such as *good – better – *goodest* is not, and this prediction seems to be cross-linguistically valid (Bobaljik 2012). Here, **ABA* refers to syncretism, as in Caha 2009, rather than to root suppletion (cf. Zompi 2019).

be that PAR is more marked than *both* ACC and GEN, leading to the total order in (85), summarized in (90).

(90) NOM < ACC < GEN < PAR < obliques (cf. Caha 2009: 115)

Crucially, this conclusion is unavoidable only on the assumption – made explicitly throughout Caha 2009 – that case hierarchy is a total order. However, this assumption has been questioned in subsequent literature, e.g., in Graf 2019, Zompì 2019, and Bány 2021 (see also Harðarson 2016 and Christopoulos and Zompì 2022). For example, Bány (2021) argues for the following partially ordered case hierarchy for Spanish:

(91) NOM < ACC < {GEN § DAT} < ... (Bány 2021, Spanish)

Moreover, cross-linguistic considerations show that the genitive is a relatively volatile case, exhibiting different relations to accusative and dative in different languages (Harðarson 2016, Zompì 2019, Starke 2017, Bány 2021). The partially ordered case hierarchy that we proposed above, repeated below as (92), is compatible with these considerations.

(92) NOM < ACC < {GEN § PAR} < obliques (Estonian)

Note that this partial hierarchy satisfies the uncontroversial relations $NOM <^1 ACC <^1 GEN$ argued for in Caha 2009. Moreover, it is also compatible with the observation that PAR.SG forms such as *aasta-t* and *kirik-u-t* in table (89) contain the GEN.SG=ACC.SG forms such *aasta* and *kirik-u*. However, the explanation here is different than in Caha 2009: this containment is conditioned by the relation $ACC < PAR$ in (92), not by the hypothetical $GEN < PAR$ postulated by Caha.⁸¹

In partial summary, both case hierarchies are consistent with the Estonian syncretism and containment data discussed in Caha 2009: the partial order in (92), which is compatible with our case-stacking analysis of Estonian pseudopartitives, and Caha’s total order in (90), which is not fully compatible with this analysis. But are there independent arguments that would make it possible to choose between these two options?

In fact, there is a rather direct argument for the partially ordered Estonian case hierarchy. As often observed, Estonian oblique cases – illative, inessive, etc. – are based on the genitive, both in the singular and in plural. This can be observed in table (78), as well as in the tables below, which extend the above tables (88)–(89) with three oblique cases: illative, allative, and translative (the others are formed fully regularly, as in (78)).

(93) Estonian cases (structural and some oblique), plural

position	‘flag.PL’	‘church.PL’	‘person.PL’
NOM	lipud	kirikud	inimesed
ACC	lipud	kirikud	inimesed
GEN	lippude	kirikute	inimeste
PAR	lippusid	kirikusid	inimesi
ILL	lippude-sse	kirikute-sse	inimeste-sse
ALL	lippude-le	kirikute-le	inimeste-le
TRL	lippude-ks	kirikute-ks	inimeste-ks

(94) Estonian cases (structural and some oblique), singular

position	‘corridor.SG’	‘person.SG’	‘year.SG’	‘church.SG’
NOM	koridor	inimene	aasta	kirik
ACC	koridor-i	inimes-e	aasta	kirik-u
GEN	koridor-i	inimes-e	aasta	kirik-u
PAR	koridor-i	inimes-t	aasta-t	kirik-u-t
ILL	koridor-i-sse	inimes-e-sse	aasta-sse	kirik-u-sse
ALL	koridor-i-le	inimes-e-le	aasta-le	kirik-u-le
TRL	koridor-i-ks	inimes-e-ks	aasta-ks	kirik-u-ks

Now, the argument against the total order “... < GEN < PAR < obliques” is based on the observation that the fully regular containment of GEN within oblique cases would usually have to skip the intervening PAR, violating the *ABA generalization. This is fully clear in the case of plural forms in table (93), where for example *lippudesse* ‘flag.ILL.PL’ contains the genitive *lippude*, and not the partitive *lippusid*, as might be expected given Caha’s (2009) total order. Similarly, in the case of singular forms, the same argument can be made on the basis of those paradigms in which GEN ≠ PAR, especially where neither is contained in the other, as in the ‘person.SG’ column.

By contrast, these data are not problematic on the partially ordered case hierarchy assumed above, as PAR does not intervene between GEN and the oblique cases. Since PAR and GEN are not comparable, the oblique cases are immediately more marked than both GEN and PAR, and they happen to be morphologically based on the former.

⁸¹ Additional data discussed in Caha 2009: 130–134 may be reanalyzed the same way.

In fact, the frequently mentioned generalization that oblique cases are uniformly based on the genitive is only an approximation, and the full range of data provides further support for the partially-ordered case hierarchy. The crucial observation is that, in brief, while oblique cases are normally based on the genitive, they are sometimes based on the partitive. This is not surprising if Estonian obliques are immediately more marked than *both* PAR and GEN, as proposed above.

There are two situations in which Estonian obliques may be based on the partitive rather than the genitive. First, in the singular, the regular “long” illative formed with *-sse* has an alternative “short” form. This short illative may have different forms, but there is a class of roots for which it is identical to the partitive, e.g.:⁸²

(95) Estonian singular illative (some lexemes)

position	‘bird.sg’		‘hoof.sg’		‘smith.sg’	
NOM	lind		kabi		sepp	
GEN	linnu		kabja		sepa	
PAR		lindu		kapja		seppa
ILL	linnu-sse	lindu	kabja-sse	kapja	sepa-sse	seppa
ALL	linnu-le		kabja-le		sepa-le	
TRL	linnu-ks		kabja-ks		sepa-ks	

As convincingly argued in Hirvonen 2020, short illative is a form of illative, rather than a separate case, as sometimes assumed: it occurs in the same positions as the regular illative, short and long illative forms of nouns and adjectives may co-occur within a single nominal phrase, etc.⁸³ Such data are readily compatible with the view, defended above, that Estonian oblique cases are immediately more marked than *both* GEN and PAR, but less clearly compatible with the totally-ordered case hierarchy, on which Estonian obliques would be immediately more marked than *only* GEN.⁸⁴

While in the singular such “parallel forms” (as they are called in Estonian linguistics) are only possible in ILL, with other obliques uniformly based on the genitive (see ALL and TRL in table (95)), in the plural some lexemes allow for similarly parallel forms of all true obliques, and when they do, these additional forms are sometimes based on PAR, as illustrated in the following table:

(96) Estonian plural obliques (some lexemes)

position	‘person.PL’		‘corridor.PL’	
NOM	inimesed		koridorid	
GEN	inimeste		koridoride	
PAR		inimesi		koridore
ILL	inimeste-sse	inimesi-sse	koridoride-sse	koridore-sse
INE	inimeste-s	inimesi-s	koridoride-s	koridore-s
ELA	inimeste-st	inimesi-st	koridoride-st	koridore-st
ALL	inimeste-le	inimesi-le	koridoride-le	koridore-le
ADE	inimeste-l	inimesi-l	koridoride-l	koridore-l
ABL	inimeste-lt	inimesi-lt	koridoride-lt	koridore-lt
TRL	inimeste-ks	inimesi-ks	koridoride-ks	koridore-ks

Again, such effects are expected if oblique cases are immediately more marked than *both* PAR and GEN, but surprising if one of these two cases intervenes between the other and the obliques.

In summary, the partial order of cases needed to account for the heterogeneous vs. homogeneous pattern in Estonian pseudopartitives has an independent – morphological – motivation and, hence, it satisfies Norris’s (2018) objective “to avoid a stipulated case hierarchy”.

6.5 Comparison to Norris 2018

As noted above, Norris (2018) rejects an analysis based on case stacking and exponence rules. Instead, he offers an account based on the *Case Concord* principle in (97) and the *Unmarked Case* principle in (98):

⁸²Accusative is omitted in (95)–(96); recall that ACC.SG=GEN.SG and ACC.PL=NOM.PL.

⁸³Moreover, the variation between short and long forms is limited: for any lexeme, one of the forms, usually the short illative, is clearly preferred by native speakers and much more frequent than the other form in corpora.

⁸⁴However, these data may also be analyzed in purely phonological terms, as in Hirvonen 2020, which weakens the current argument. The next argument, based on plural forms of obliques, does not seem amenable to such a purely phonological analysis.

(97) *Case Concord*:

Let X and Y be two nodes in a single extended projection, Y immediately dominating X.

If Y has a valued case feature [CASE: α] (but X does not), then copy Y's case feature to X.

(98) *Unmarked Case* (Estonian KP):

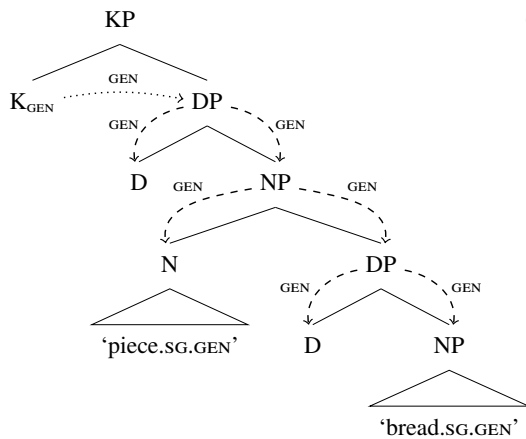
Assign partitive case to complements of N^0 that do not already have a case value when the pseudopartitive extended projection is complete.

Case Concord ensures that a case feature on a node percolates downwards. It is parameterized: it may hold in some languages but not in others, and when it does hold, case may percolate to nodes that already have case (accounting for overt case stacking in some languages) or not (accounting for languages such as Estonian). Note also that this principle only accounts for case concord, not for concord in ϕ features.

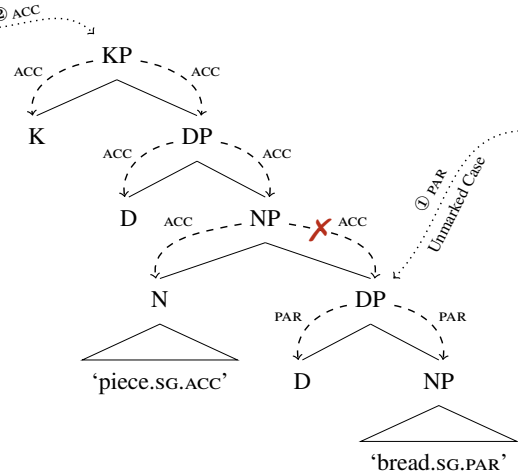
Similarly, *Unmarked Case* depends on language (here, Estonian) and maximal extended projection (here, KP). This principle is unusual in the sense that it seems to boil down to nominal heads assigning the partitive case to their complements, but unlike other such assignments, this one is context-sensitive: it happens only once the maximal extended projection of this nominal head – a KP – is constructed. For this reason it is packaged as a rule responsible for assigning the unmarked case in the sense of Dependent Case Theory (DCT; Marantz 1991, Bobaljik 2008, Baker 2015), but unlike other such rules postulated in the DCT literature, this one is limited to a specific constituent in a specific position, rather than any case-less nominal projection within a given domain.

The analysis of Estonian pseudopartitives is based on the additional assumption that when K merges with DP, it may assign a specific oblique case such as GEN or ILL, or it may be unspecified for case, in which case it receives NOM or ACC later on in the derivation via DCT principles, depending on the position of the KP. If K assigns a specific case, then this case percolates down according to the *Case Concord* principle before the *Unmarked Case* has a chance to mark the relevant DP with PAR; this is illustrated in (99) below, with dashed lines indicating case copying effects of *Case Concord*. On the other hand, when K is not marked for a specific case, then the *Unmarked Case* causes the DP to PAR. As illustrated in (100) below, this stops the percolation of the NOM or ACC assigned later on to the whole KP, as Estonian is parameterized for the presence of the “but X does not” bit of the *Case Concord* principle in (97).

(99)



(100)



We claim that the approach to concord advocated in previous sections is theoretically leaner. Where the approach in Norris 2018 postulates two syntactic principles, one of which is specific to case concord (with other principles needed for concord in ϕ features; cf. Norris 2014) and the other rather ad-hoc, the current approach is based on independently motivated and commonly assumed syntactic mechanisms (Agree, phases), with the one new morphological principle – *Exponere Outer Most Marked* – generalizing exponence rules of stacked feature bundles proposed previously (e.g., Baker and Vinokurova 2010, Richards 2013, Pesetsky 2013, Ionin and Matushansky 2018: §6.3), also outside the context of nominal concord (e.g., Clem and Deal 2024, Grishin 2024). Moreover, as demonstrated above and in the following subsection, cross-linguistic variation on the current approach is a matter of properties of functional and semi-lexical heads, which is more in line with the Borer-Chomsky Conjecture⁸⁵ than the two parameterizable principles in (97)–(98).

Arguably, the current approach is also superior empirically. For example, the initially puzzling constraints on overt

⁸⁵As formulated in Baker 2008: 156, (3):

(i) *The Borer-Chomsky Conjecture*

All parameters of variation are attributable to differences in the features of particular items (e.g., the functional heads) in the lexicon.

case stacking discussed in §3.5, where up to two overt cases resulting from concord are observed, fall out from the current approach without further ado. By contrast, on the approach of Norris 2018, this constraint is not automatically predicted (although perhaps it could be modeled by making the *Case Concord* principle in (97) sensitive to phases).

Moreover, as case concord is dissociated from ϕ concord, it is not clear how to translate to Norris’s (2018) framework the analysis of Russian paucals in §5.2.2, where the counted noun in the genitive bears the *paucal* number in NOM and ACC positions but the *plural* number in GEN. This is unexpected on the approach in Norris 2014, 2018, where separate principles are responsible for case and ϕ features in nominal concord. By contrast, the current approach handles such data relatively easily and does not rely on specialized concord rules.

The following section demonstrates that also the cross-linguistic data presented in Norris 2018 as “promising” for the approach to concord presented there falls out naturally from the approach proposed in previous sections.

6.6 Finnish and Other Languages

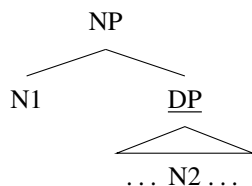
As discussed in Norris 2018: §5.2 on the basis of data mainly from Brattico 2008, 2010, 2011, there is an interesting contrast between Estonian and the closely related Finnish: while Estonian pseudopartitives show the heterogeneous vs. homogeneous pattern discussed above, Finnish displays a strictly heterogeneous pattern, with N2 always in the partitive, even in oblique positions, e.g. (Norris 2018: 554, (65b), (64b)):

- (101) selle-st hulga-st auto-de-st (EST)
 this-ELA group-ELA car-PL-ELA
 ‘from this group of cars’
- (102) siitä jouko-sta auto-ja (FIN)
 that.ELA set-ELA car-PL.PAR
 ‘from that set of cars’

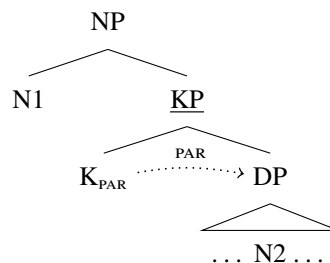
This is a little surprising, as in both languages the same heterogeneous vs. homogeneous pattern is witnessed in numeral phrases: the counted noun is in the partitive in NOM and ACC positions, and in the appropriate agreeing case in other positions.

Norris 2018: 555 analyzes the difference illustrated in (101)–(102) as follows:

(103) Estonian pseudopartitives:



(104) Finnish pseudopartitives:



First, the two languages differ in the structure of pseudopartitives: the complement of the N1 is a DP in Estonian but a KP in Finnish. Second, the partitive in Finnish pseudopartitives is inherent, unlike the unmarked partitive in Estonian pseudopartitives and in both Finnish and Estonian numeral phrases. This second assumption is needed to ensure that the “but X does not” bit of the *Case Concord* principle in (97) blocks in Finnish the downward percolation of whatever case the whole pseudopartitive construction receives.

Note that on the current analysis only the first of these two differences between Estonian and Finnish needs to be assumed: since the complement of N1 is a KP, no constituent within this KP may be flagged with an external case by virtue of standard assumptions about phase boundaries.

Norris (2018: §5.3) considers also some cross-linguistic variation in the marking of the N2 in pseudopartitives, suggesting that it results from the fact that languages are parameterized for the presence or absence of each of the two principles in (97)–(98), as shown in the following table:

(105) Marking of the N2 phrase in pseudopartitives (Norris 2018: 558, Table 3):

		<i>Unmarked Case</i>	
		+	–
<i>Case Concord</i>	+	Estonian	German, Greek
	–	English	Danish

Thus, Estonian has both *Case Concord* (CC) and *Unmarked Case* (UC), German and Greek have CC but not UC, English has UC (realized as the preposing *of*) but not CC, while Danish has neither:

- (106) zwei Gläser Wasser (DEU; +CC, –UC)
 two glasses.NOM water.NOM

- (107) to glas vand (DAN; –CC, –UC)
 two glasses water
 ‘two glasses of water’ (ENG; –CC, +UC)

On the current approach, this variation is explained equally readily, without any specialized concord principles; it is sufficient to replace $\pm$ Case Concord in (105) with $\pm$ morphological case, and $\pm$ Unmarked Case with $\pm$ case flagging. Thus, in German and Greek, which – just like Estonian – have morphological case on lexical nouns, N1 does not assign any case to its DP complement, so the case assigned by the higher K does not compete with any other case on N1 and, even when that case is the most unmarked NOM or ACC, it may be realized overtly (see (106)). By contrast, English and Danish do not have nominal morphological case, and even their pronominal case systems are occasionally analyzed as plain allomorphy (see, e.g., Hudson 1995, as well as Weisser 2020 and references therein), so no case suffixes are observed there (see (107)). The difference between English and Dutch is that the complement of N1 is marked with *of* in English, but unmarked in Dutch – a fact that is easy to model by appropriate properties of the English semi-lexical N1.⁸⁶

6.7 Interim Summary

In summary, Finnic data confirm the analysis of concord as insatiable agree proposed in previous sections. The difference between Estonian and Finnic may be analyzed as in Norris 2018, in terms of different selectional properties of N1, but unlike in the analysis of Norris 2018, no additional assumptions about the different status of the partitive in Estonian and Finnish are necessary. Moreover, the heterogeneous vs. homogeneous pattern in Estonian pseudopartitives is explained by exactly the same case stacking and exponence rules as similar patterns in Slavic numerals, once the partial order of Estonian case hierarchy has been established on independent morphological grounds. The fact that the *Expone Outer Most Marked* rule proposed on the basis of Russian paucals is also needed to account for Estonian pseudopartitives suggests that this approach to exponence may be cross-linguistically valid.

This last conclusion is supported by the fact that this rule seems to have the desired effect in many analyses where other exponence rules have been proposed. First, it accounts for the Russian facts for which the *Expone Outermost* was invoked (Pesetsky 2013). Second, it gives the right result in various places in which a rule like *Expone Most Marked* was used, again including Russian numerals (Matushansky 2008, Ionin and Matushansky 2018), but also Gothic free relatives (Grishin 2024). Finally, in their analysis of Sakha, Baker and Vinokurova 2010 invoke a rule that could be called *Expone Innermost*, in order to account for the dative case on nominals first assigned DAT and then ACC, i.e., with the DAT.ACC stack. Given that nominals with the NOM.ACC stack expone the outermost accusative, they need to add a special proviso to this rule, to the effect that morphologically unmarked cases, such as NOM, do not count when determining the innermost case. By contrast, both the *Expone Most Marked* rule and our *Expone Outer Most Marked* variant predict that DAT.ACC is expounded as DAT and NOM.ACC as ACC, given the commonly assumed markedness order NOM < ACC < DAT (Blake 2001, Caha 2009, Smith *et al.* 2019, Zoppi 2019, 2023).

7 Conclusion

To be done later.

References

- Steven ABNEY (1987), *The English Noun Phrase in its sentential aspect*, Ph.D. dissertation, Massachusetts Institute of Technology, Cambridge, MA.
- Sarah ASINARI (2019), Case percolation in Russian numeral constructions, *University of Pennsylvania Working Papers in Linguistics*, 25(1):9–20, URL <https://repository.upenn.edu/handle/20.500.14332/45238>, proceedings of the 42nd Annual Penn Linguistics Conference.
- Peter AUSTIN (1981), *A grammar of Diyari, South Australia*, Cambridge University Press, Cambridge.
- Leonard H. BABBY (1975), *A transformational grammar of Russian adjectives*, Mouton, The Hague / Paris.
- Leonard [H.] BABBY (1985), Prepositional quantifiers and the direct case condition in Russian, in Michael S. FLIER and Richard D. BRECHT, editors, *Issues in Russian Morphosyntax*, pp. 91–117, Slavica Publishers, Columbus, OH.
- Leonard H. BABBY (1987), Case, pre-quantifiers, and discontinuous agreement in Russian, *Natural Language and Linguistic Theory*, 5:91–138.
- John F. BAILYN and Andrew NEVINS (2008), Russian genitive plurals are impostors, in Asaf BACHRACH and Andrew NEVINS, editors, *Inflectional Identity*, pp. 237–270, Oxford University Press, Oxford.
- Mark C. BAKER (2008), *The syntax of agreement and concord*, Cambridge University Press, Cambridge.

⁸⁶If *of* is analyzed as a preposition, these would be selectional properties of N1. If it is analyzed as a phrasal case marker, these would be case-flagging properties of N1, as suggested in Norris 2018: 557 and in the main text. We remain agnostic on this issue, noting only that either analysis is immediate and does not require specialized concord principles.

- Mark C. BAKER (2015), *Case: Its principles and its parameters*, Cambridge University Press, Cambridge.
- Mark C. BAKER and Nadezhda VINOKUROVA (2010), Two modalities of case assignment: Case in Sakha, *Natural Language and Linguistic Theory*, 28:593–642, doi:<https://doi.org/10.1007/s11049-010-9105-1>.
- İsa Kerem BAYIRLI (2017), *The universality of concord*, Ph.D. dissertation, Massachusetts Institute of Technology, Cambridge, MA.
- Maria BITTNER (1994), *Case, scope, and binding*, Springer.
- Maria BITTNER and Ken HALE (1996a), Ergativity: Toward a theory of a heterogeneous class, *Linguistic Inquiry*, 27(4):531–604.
- Maria BITTNER and Ken HALE (1996b), The structural determination of case and agreement, *Linguistic Inquiry*, 27(1):1–68.
- Barry J. BLAKE (2001), *Case*, Cambridge University Press, Cambridge, 2nd edition.
- Jonathan David BOBALJIK (2008), Where’s phi? Agreement as a postsyntactic operation, in Harbour *et al.* (2008), pp. 295–328.
- Jonathan David BOBALJIK (2012), *Universals in comparative morphology: Suppletion, superlatives, and the structure of words*, MIT Press, Cambridge, MA.
- Jonathan David BOBALJIK and Uli SAUERLAND (2018), *ABA and the combinatorics of morphological features, *Glossa: A Journal of General Linguistics*, 3:1–34, doi:<https://doi.org/10.5334/gjgl.345>.
- Corentin BOURDEAU (2015), *Ergativity in Shawi (Chayahuita)*, Master’s thesis, Radboud University, Nijmegen.
- Željko BOŠKOVIĆ (2014), Now I’m a phase, now I’m not a phase: On the variability of phases with extraction and ellipsis, *Linguistic Inquiry*, 45(1):27–89.
- Pauli BRATTICO (2008), Kayne’s model of case and Finnish nominal phrases, *Nordic Journal of Linguistics*, 31:135–160.
- Pauli BRATTICO (2010), The two-part models and one-part models of nominal case: Evidence from case distribution, *Journal of Linguistics*, 45:47–81.
- Pauli BRATTICO (2011), Case assignment, case concord, and the quantificational case construction, *Lingua*, 121:1042–1066.
- Joan BRESNAN, Ash ASUDEH, Ida TOIVONEN, and Stephen WECHSLER (2016), *Lexical-functional syntax*, Blackwell Textbooks in Linguistics, Wiley-Blackwell, 2nd edition.
- Walter BREU (2020), Paucal, in Marc L. GREENBERG, editor, *Encyclopedia of Slavic Languages and Linguistics Online*, Brill, doi:http://dx.doi.org/10.1163/2589-6229_ESLO_COM_031985.
- Benjamin BRUENING (2023), Selectional violations in coordination (a response to Patejuk and Przepiórkowski 2023), doi:https://doi.org/10.1162/ling_a_00506, URL https://direct.mit.edu/ling/article-abstract/doi/10.1162/ling_a_00506/115205/Selectional-Violations-in-Coordination-A-Response, to appear in *Linguistic Inquiry*.
- Benjamin BRUENING and Eman AL KHALAF (2020), Category mismatches in coordination revisited, *Linguistic Inquiry*, 51(1):1–36, doi:10.1162/ling_a_00336.
- András BÁRÁNY (2021), Partially ordered case hierarchies, *Glossa: A Journal of General Linguistics*, 6:1–19, doi:<https://doi.org/10.5334/gjgl.1451>.
- Susana BÉJAR and Milan ŘEZÁČ (2009), Cyclic agree, *Linguistic Inquiry*, 40(1):35–73, doi:<https://doi.org/10.1162/ling.2009.40.1.35>.
- Pavel CAHA (2009), *The nanosyntax of case*, Ph.D. dissertation, University of Tromsø.
- Pavel CAHA (2024), A default theory of default case, *Glossa: A Journal of General Linguistics*, 9:1–43, doi:<https://doi.org/10.16995/glossa.16571>.
- Tingchun CHEN (2018), *Multiple case assignment: An Amis case study*, Ph.D. dissertation, Massachusetts Institute of Technology, Cambridge, MA.
- Noam CHOMSKY (1995), Bare phrase structure, in Gert WEBELHUTH, editor, *Government and Binding Theory and the Minimalist Program*, number 1 in Generative Syntax, pp. 383–439, Blackwell, Oxford.
- Noam CHOMSKY (2000), Minimalist inquiries: The framework, in Roger MARTIN, David MICHAELS, and Juan URIAGEREKA, editors, *Step by Step*, pp. 89–155, MIT Press, Cambridge, MA.
- Noam CHOMSKY (2001), Derivation by phase, in Michael KENSTOWICZ, editor, *Ken Hale: A Life in Language*, pp. 1–52, MIT Press, Cambridge, MA, doi:<https://doi.org/10.7551/mitpress/4056.003.0004>.
- Noam CHOMSKY (2007), Approaching UG from below, in Uli SAUERLAND and Hans-Martin GÄRTNER, editors, *Interfaces + Recursion = Language? Chomsky’s Minimalism and the View from Syntax-Semantics*, pp. 1–29, Walter de Gruyter, Berlin.
- Noam CHOMSKY (2008), On phases, in Robert FREIDIN, Carlos P. OTERO, and Maria Luisa ZUBIZARRETA, editors, *Foundational Issues in Linguistic Theory: Essays in Honor of Jean-Roger Vergnaud*, pp. 133–166, MIT Press, Cambridge, MA.

- Christos CHRISTOPOULOS and Stanislaw ZOMPI (2022), Taking the nominative (back) out of the accusative, *Natural Language and Linguistic Theory*, 41(3):879–909, doi:<https://doi.org/10.1007/s11049-022-09557-y>.
- Barbara CITKO (2000), *Parallel merge and the syntax of free relatives*, Ph.D. dissertation, State University of New York at Stony Brook.
- Barbara CITKO (2005), On the nature of merge: External merge, internal merge, and parallel merge, *Linguistic Inquiry*, 36(4):475–496.
- Barbara CITKO (2011), *Symmetry in syntax: Merge, move and labels*, Cambridge University Press, Cambridge.
- Barbara CITKO (2014), *Phase theory: An introduction*, Cambridge University Press, Cambridge.
- Barbara CITKO, Allison GERMAIN, and Jacek WITKOŚ (2018), If you cannot agree, move on! On labels and non-nominative subjects, *Glossa: A Journal of General Linguistics*, 3:1–46, doi:<https://doi.org/10.5334/gjgl.399>.
- Barbara CITKO and Martina GRAČANIN-YÜKSEK (2020), *Merge: Binariness in (multidominant) syntax*, MIT Press, Cambridge, MA.
- Emily CLEM (2019), Amahuaca ergative as agreement with multiple heads, *Natural Language and Linguistic Theory*, 37(3):785–823.
- Emily CLEM (2023), Cyclic expansion in agree: Maximal projections as probes, *Linguistic Inquiry*, 54(1):39–78.
- Emily CLEM and Amy Rose DEAL (2024), Dependent case by agree: Ergative in Shawi, doi:https://doi.org/10.1162/ling_a_00529, to appear in *Linguistic Inquiry*, https://direct.mit.edu/ling/article-abstract/doi/10.1162/ling_a_00529/119973/Dependent-Case-by-Agree-Ergative-in-Shawi.
- Greville G. CORBETT (1978), Numerous squishes and squishy numerals in Slavonic, *International Review of Slavic Linguistics*, 3(1–2):43–73.
- Greville G. CORBETT (1993), The head of Russian numeral expressions, in Greville G. CORBETT, Norman M. FRASER, and Scott MCGLASHAN, editors, *Heads in Grammatical Theory*, pp. 11–35, Cambridge University Press, Cambridge.
- Greville G. CORBETT (2000), *Number*, Cambridge University Press, Cambridge.
- Mary DALRYMPLE, editor (2023), *Handbook of Lexical Functional Grammar*, Language Science Press, Berlin, URL <https://langsci-press.org/catalog/book/312>.
- Mary DALRYMPLE, John J. LOWE, and Louise MYCOCK (2019), *The Oxford reference guide to Lexical Functional Grammar*, Oxford University Press, Oxford, doi:<https://doi.org/10.1093/oso/9780198733300.001.0001>.
- Amy Rose DEAL (2010), Ergative case and the transitive subject: A view from Nez Perce, *Natural Language and Linguistic Theory*, 28(1):73–120.
- Amy Rose DEAL (2015), Interaction and satisfaction in ϕ -agreement, in Thuy BUI and Deniz ÖZYILDIZ, editors, *Proceedings of the Forty-Fifth Annual Meeting of the North East Linguistic Society*, pp. 179–192, Amherst, MA, URL <https://ling.auf.net/lingbuzz/002610/>.
- Amy Rose DEAL (2023), Current models of Agree, URL <https://ling.auf.net/lingbuzz/006504/>, to appear in James Crippen, Rose-Marie Dechaine and Hermann Keupdjio (eds.), *Move and Agree: Towards a formal typology*. John Benjamins. <https://ling.auf.net/lingbuzz/006504/>, version of March 2023.
- Amy Rose DEAL (2024), Interaction, satisfaction, and the PCC, *Linguistic Inquiry*, 55(1):39–94, doi:https://doi.org/10.1162/ling_a_00455.
- Katarzyna DZIWIŃSKA (1990), Default agreement in Polish, in Katarzyna DZIWIŃSKA, Patrick FARRELL, and Errapel MEJÍAS-BIKANDI, editors, *Grammatical Relations: A Cross-Theoretical Perspective*, pp. 147–161, CSLI Publications, Stanford, CA.
- Nicholas D. EVANS (1995), Multiple case in Kayardild: Anti-iconic suffix ordering and the diachronic filter, in Plank (1995), pp. 396–428.
- Steven FRANKS (1994), Parametric properties of numeral phrases in Slavic, *Natural Language and Linguistic Theory*, 12(4):597–674.
- Steven FRANKS (1995), *Parameters of Slavic morphosyntax*, Oxford University Press, New York.
- Steven L. FRANKS (2018), A Bulgarian solution to the Slavic Q question?, in Franks *et al.* (2018), pp. 93–120.
- Steven L. FRANKS, Vrinda CHIDAMBARAM, Brian D. JOSEPH, and Iliyana KRAPOVA, editors (2018), *Katerino mome: Studies in Bulgarian morphosyntax in honor of Catherine Rudin*, Slavica Publishers, Bloomington, IN.
- Thomas GRAF (2019), Monotonicity as an effective theory of morphosyntactic variation, *Journal of Language Modelling*, 7(2):3–47, doi:<https://doi.org/10.15398/jlm.v7i2.211>, URL <https://jlm.ipipan.waw.pl/index.php/JLM/article/view/211>.
- Jane GRIMSHAW (2000), Locality and extended projection, in Peter COOPMANS, Martin EVERAERT, and Jane GRIMSHAW, editors, *Lexical Specification and Insertion*, pp. 115–133, John Benjamins, Amsterdam/Philadelphia.
- Peter GRISHIN (2024), How to agree with the lowest DP, <https://ling.auf.net/lingbuzz/007112>, version of May

2024.

- Christopher Mathias HAMMERLY (2020), *Person-based prominence in Ojibwe*, Ph.D. dissertation, University of Massachusetts Amherst.
- Wayne HARBERT (1992), Gothic relative clauses and syntactic theory, in Irmengard RAUCH, Gerald F. CARR, and Robert L. KYES, editors, *On Germanic Linguistics: Issues and Methods*, Mouton de Gruyter, Berlin/New York.
- Daniel HARBOUR, David ADGER, and Susana BÉJAR, editors (2008), *Phi theory: Phi-features across modules and interfaces*, Oxford University Press, Oxford.
- Heidi HARLEY and Elizabeth RITTER (2002), Person and number in pronouns: A feature-geometric analysis, *Language*, 78(3):482–526.
- Gísli Rúnar HARÐARSON (2016), A case for a Weak Case Contiguity hypothesis—a reply to Caha, *Natural Language and Linguistic Theory*, 34(4):1329–1343, doi:<https://doi.org/10.1007/s11049-016-9328-x>.
- Johannes HIRVONEN (2020), *The Estonian illative: Productivity and formalization*, Master's thesis, Universität Wien, Vienna.
- Richard HUDSON (1987), Zwicky on heads, *Journal of Linguistics*, 23(1):109–132.
- Richard HUDSON (1995), Does English really have case?, *Journal of Linguistics*, 31(2):375–392.
- Tania IONIN and Ora MATUSHANSKY (2018), *Cardinals: The syntax and semantics of cardinal-containing expressions*, MIT Press, Cambridge, MA.
- Ronald M. KAPLAN (1987), Three seductions of computational psycholinguistics, in P. WHITELOCK, M. M. WOOD, H. L. SOMERS, R. JOHNSON, and P. BENNETT, editors, *Linguistic Theory and Computer Applications*, pp. 149–188, Academic Press, London.
- Ronald M. KAPLAN and Joan BRESNAN (1982), Lexical-Functional Grammar: A formal system for grammatical representation, in Joan BRESNAN, editor, *The Mental Representation of Grammatical Relations*, pp. 173–281, MIT Press, Cambridge, MA.
- Stefan KEINE (2019), Selective opacity, *Linguistic Inquiry*, 50(1):13–61, doi:https://doi.org/10.1162/ling_a_00299.
- Stefan KEINE (2020), *Probes and their horizons*, MIT Press, Cambridge, MA, doi:<https://doi.org/10.7551/mitpress/12003.001.0001>.
- Heidi KLOCKMANN (2017), *The design of semi-lexicity: Evidence from case and agreement in the nominal domain*, Ph.D. dissertation, Utrecht University.
- Greg LAMONTAGNE and Lisa TRAVIS (1987), The syntax of adjacency, in Megan CROWHURST, editor, *Proceedings of the Sixth West Coast Conference on Formal Linguistics*, pp. 173–186, CSLI Publications, Stanford, CA.
- Theodore LEVIN (2015), *Licensing without case*, Ph.D. dissertation, Massachusetts Institute of Technology, Cambridge, MA.
- Theodore LEVIN (2017), Successive-cyclic case assignment: Korean nominative-nominative case-stacking, *Natural Language and Linguistic Theory*, 35(2):447–498.
- Alec MARANTZ (1991), Case and licensing, in German WESTPHAL, Benjamin AO, and Hee-Rahk CHAE, editors, *8th Eastern States Conference on Linguistics (ESCOL 91)*, pp. 234–253, CLC Publications, Ithaca, NY.
- Ora MATUSHANSKY (2008), A case study of predication, in Franc MARUŠIČ and Rok ŽAUCER, editors, *Studies in Formal Slavic Linguistics: Contributions from Formal Description of Slavic Languages 6.5*, pp. 213–239, Peter Lang, Frankfurt am Main.
- Witold MAŃCZAK (1956), Ile jest rodzajów w polskim?, *Język Polski*, 36(2):116–121.
- Thomas McFADDEN (2018), *ABA in stem-allomorphy and the emptiness of the nominative, *Glossa: A Journal of General Linguistics*, 3:1–36, doi:<https://doi.org/10.5334/gjgl.373>.
- Igor MEL'ČUK (1985), *Poverkhnostnyj sintaksis russkix čislovyx vyraženij*, Gesellschaft zur Förderung Slawistischer Studien, Vienna.
- Katarzyna MIECHOWICZ-MATHIASSEN (2012), Licensing Polish higher numerals: An account of the accusative hypothesis, in Joanna BŁASZCZAK, Bożena ROZWADOWSKA, and Wojciech WITKOWSKI, editors, *Current Issues in Generative Linguistics: Syntax, Semantics and Phonology*, volume 2 of *Generative Linguistics in Wrocław*, pp. 58–75, University of Wrocław, Wrocław.
- Edith A. MORAVICSIK (1995), Summing up Suffixaufnahme, in Plank (1995), pp. 451–484.
- Beata MOSKAL (2015), *Domains on the border: Between morphology and phonology*, Ph.D. dissertation, University of Connecticut, URL <https://opencommons.uconn.edu/dissertations/892>.
- Gereon MÜLLER (2007), Extended exponence by enrichment: Argument encoding in German, Archi, and Timucua, *University of Pennsylvania Working Papers in Linguistics*, 13(1):9–20, URL <https://repository.upenn.edu/handle/20.500.14332/44619>, proceedings of the 30th Annual Penn Linguistics Colloquium.

- Stefan MÜLLER, Anne ABEILLÉ, Robert D. BORSLEY, and Jean-Pierre KOENIG, editors (2024), *Head-driven Phrase Structure Grammar: The handbook*, Language Science Press, Berlin, 2nd edition, URL <https://langsci-press.org/catalog/book/478>.
- Tore NESSET (2020), A long birth: The development of gender-specific paucal constructions in Russian, *Diachronica*, 37(4):514–539.
- Tore NESSET and Maria NORDRUM (2019), Do Russian paucal numerals govern the genitive? Evidence from stress placement, *Russian Linguistics*, 43(2):87–105.
- Andrew NEVINS (2011), Marked targets versus marked triggers and impoverishment of the dual, *Linguistic Inquiry*, 42(3):413–444, doi:https://doi.org/10.1162/ling_a_00052.
- Mark NORRIS (2014), *A theory of nominal concord*, Ph.D. dissertation, University of California Santa Cruz.
- Mark NORRIS (2018), Unmarked case in Estonian nominals, *Natural Language and Linguistic Theory*, 36(2):523–562.
- Mark NORRIS (2022), Nominal inflection in Distributed Morphology, URL <https://ling.auf.net/lingbuzz/006516/>, to appear in Cambridge Handbook of Distributed Morphology. <https://ling.auf.net/lingbuzz/006516/>, version of March 2022.
- Roumyana PANCHEVA (2018), How many flowers! so many colors! number marking in cardinality exclamatives in Bulgarian, in Franks *et al.* (2018), pp. 197–233.
- Agnieszka PATEJUK and Adam PRZEPIÓRKOWSKI (2018), Predicative constructions with infinitival and clausal subjects, in Miriam BUTT and Tracy Holloway KING, editors, *The Proceedings of the LFG'18 Conference*, pp. 304–324, CSLI Publications, Stanford, CA, URL <https://web.stanford.edu/group/cslipublications/cslipublications/LFG/LFG-2018/>.
- Agnieszka PATEJUK and Adam PRZEPIÓRKOWSKI (2021), Predicative adverbs: Evidence from Polish, *Linguistic Inquiry*, 52(4):835–851, doi:https://doi.org/10.1162/ling_a_00394, URL https://www.mitpressjournals.org/doi/abs/10.1162/ling_a_00394.
- Agnieszka PATEJUK and Adam PRZEPIÓRKOWSKI (2023), Category mismatches in coordination vindicated, *Linguistic Inquiry*, 54(2):326–349, doi:https://doi.org/10.1162/ling_a_00438.
- Asya PERELTSVAIG (2010), As easy as two, three, four?, in Wayles BROWNE, Adam COOPER, Alisaon FISHER, Esra KESICI, Nikola PREDOLAC, and Draga ZEC, editors, *Annual Workshop on Formal Approaches to Slavic Linguistics: The Second Cornell Meeting 2009*, pp. 418–435, Michigan Slavic Publications, Ann Arbor, MI.
- David PESETSKY (2013), *Russian case morphology and the syntactic categories*, MIT Press, Cambridge, MA.
- Tatiana PHILIPPOVA (2018), *Prepositional repercussions in Russian: Pronouns, comparatives and ellipsis*, Ph.D. dissertation, Ben-Gurion University of the Negev, Beer-Sheva.
- Frans PLANK, editor (1995), *Double case: Agreement by Suffixaufnahme*, Oxford University Press, New York.
- Carl POLLARD and Ivan A. SAG (1987), *Information-based syntax and semantics, volume 1: Fundamentals*, CSLI Publications, Stanford, CA.
- Carl POLLARD and Ivan A. SAG (1994), *Head-driven Phrase Structure Grammar*, Chicago University Press / CSLI Publications, Chicago, IL.
- Omer PREMINGER (2014), *Agreement and its failures*, MIT Press, Cambridge, MA.
- Adam PRZEPIÓRKOWSKI (1996), Case assignment in Polish: Towards an HPSG analysis, in Claire GROVER and Enric VALLDÚVÍ, editors, *Studies in HPSG*, volume 12 of *Edinburgh Working Papers in Cognitive Science*, pp. 191–228, Centre for Cognitive Science, University of Edinburgh, URL <http://nlp.ipipan.waw.pl/~adamp/Papers/1995-case/>.
- Adam PRZEPIÓRKOWSKI (1999), *Case assignment and the complement-adjunct dichotomy: A non-configurational constraint-based approach*, Ph.D. dissertation, Universität Tübingen, URL <http://nlp.ipipan.waw.pl/~adamp/Dissertation/>.
- Adam PRZEPIÓRKOWSKI (2004), O wartości przypadku podmiotów liczebnikowych, *Biuletyn Polskiego Towarzystwa Językoznawczego*, LX:133–143.
- Adam PRZEPIÓRKOWSKI, Mirosław BAŃKO, Rafał L. GÓRSKI, and Barbara LEWANDOWSKA-TOMASZCZYK, editors (2012), *Narodowy Korpus Języka Polskiego*, Wydawnictwo Naukowe PWN, Warsaw.
- Adam PRZEPIÓRKOWSKI and Agnieszka PATEJUK (2025), Prenominal adverbs, or apparent selectional violations in coordination, *Linguistic Inquiry*, Early Access:1–29, doi:https://doi.org/10.1162/ling_a_00548.
- Natalia RAKHLIN (2003), Genitive of quantification in Russian: What morphology can tell us about syntax, in Marjo VAN KOPPEN and Mark DE VOS, editors, *Proceedings of CONSOLE XI*, Student Organization of Linguistics in Europe, Leiden, the Netherlands, URL <http://www.hum2.leidenuniv.nl/pdf/luc1/sole/console11/console11-rakhl.pdf>.
- Gilbert C. RAPPAPORT (2002), Numeral phrases in Russian: A minimalist approach, *Journal of Slavic Linguistics*,

10(1/2):327–240.

- Charles REISS (2022), Priority union and feature logic in phonology, *Linguistic Inquiry*, 53(1):199–209, doi:https://doi.org/10.1162/ling_a_00400.
- Milan ŘEZÁČ (2008), Phi-agree and theta-related case, in Harbour *et al.* (2008), pp. 83–129.
- Norvin RICHARDS (2010), *Uttering trees*, MIT Press, Cambridge, MA.
- Norvin RICHARDS (2013), Lardil “case stacking” and the timing of case assignment, *Syntax*, 61(1):42–76, doi:<https://doi.org/10.1111/j.1467-9612.2012.00169.x>.
- Elizabeth RITTER (1991), Two functional categories in modern Hebrew noun phrases, in Susan ROTHSTEIN, editor, *Perspectives on Phrase Structure: Heads and Licensing*, volume 25 of *Syntax and Semantics*, pp. 37–60, Academic Press, San Diego.
- Paweł RUTKOWSKI and Kamil SZCZEGOT (2001), On the syntax of functional elements, in Adam PRZEPÍÓRKOWSKI and Piotr BAŃSKI, editors, *Generative Linguistics in Poland: Syntax and Morphosyntax*, pp. 187–196, Institute of Computer Science, Polish Academy of Sciences, Warsaw.
- Carmen SALDANA, Patrick KANAMPIU, and Jennifer CULBERTSON (2025), Gender comes first: Experimental evidence for the representation of gender closer to the noun stem than number across linguistic populations, doi:https://doi.org/10.1162/ling_a_00552, *Linguistic Inquiry* (Just Accepted).
- Zygmunt SALONI (1974), Klasyfikacja gramatyczna leksemów polskich, *Język Polski*, LIV(1–2):3–13, 93–101.
- Alexander M. SCHENKER (1971), Some remarks on Polish quantifiers, *Slavic and East European Journal*, XV(1):54–59.
- Carson T. SCHÜTZE (2001), On the nature of default case, *Syntax*, 4(3):205–238.
- Peter W. SMITH, Beata MOSKAL, Ting XU, Jungmin KANG, and Jonathan David BOBALJIK (2019), Case and number suppletion in pronouns, *Natural Language and Linguistic Theory*, 27(3):1029–1101, doi:<https://doi.org/10.1007/s11049-018-9425-0>.
- Michał STARKE (2017), Resolving (DAT = ACC) ≠ GEN, *Glossa: A Journal of General Linguistics*, 2:1–8, doi:<https://doi.org/10.5334/gjgl.408>.
- Oscar E. SWAN (2002), *A grammar of contemporary Polish*, Slavica Publishers, Bloomington, IN.
- Marek ŚWIDZIŃSKI and Magdalena DERWOJEDOWA (2004), Idiosynkrazja na przecięciu idiosynkrazji, czyli o poprzymkowości i liczebnikach, in Andrzej MOROZ and Marek WIŚNIEWSKI, editors, *Studia z gramatyki i semantyki języka polskiego*, pp. 33–42, Wydawnictwo Uniwersytetu Mikołaja Kopernika, Toruń.
- Philipp WEISSER (2020), On the symmetry of case in conjunction, *Syntax*, 23(1):42–77, doi:<https://doi.org/10.1111/synt.12188>.
- Ewa WILLIM (2015), Case distribution and ϕ -agreement with Polish Genitive of Quantification in the Feature Sharing theory of Agree, *Poznań Studies in Contemporary Linguistics*, 51(2):315–357, doi:<http://doi.org/10.1515/psicl-2015-0013>.
- Jacek WITKOŚ, Dominika DZIUBAŁA-SZREJBROWSKA, Piotr CEGŁOWSKI, and Paulina ŁĘSKA (2018), *The syntax of numeral noun constructions: A view from Polish*, Peter Lang, Berlin.
- Marcin WOLIŃSKI, Zygmunt SALONI, Robert WOŁOSZ, Włodzimierz GRUSZCZYŃSKI, Danuta SKOWROŃSKA, and Zbigniew BRONK (2020), *Słownik gramatyczny języka polskiego*, Warsaw, 4th edition, <http://sgjp.pl/> (beta).
- Michael YADROFF (1999), *Formal properties of functional categories: The minimalist syntax of Russian nominal and prepositional expressions*, Ph.D. dissertation, Indiana University.
- Andrey Anatolyevich ZALIZNJAK (1967), *Russkoe imennoe slovoizmenenie*, Nauka, Moscow.
- Stanislao ZOMPÌ (2019), Ergative is not inherent: Evidence from *ABA in suppletion and syncretism, *Glossa: A Journal of General Linguistics*, 4:1–28, doi:<https://doi.org/10.5334/gjgl.816>.
- Stanislao ZOMPÌ (2023), **ABA in multidimensional paradigms: A Max/Dep-based account*, Ph.D. dissertation, Massachusetts Institute of Technology, Cambridge, MA.
- Arnold M. ZWICKY (1985), Heads, *Journal of Linguistics*, 21(1):1–29.

A Appendix: Technicalities

A.1 Agreement and Flagging

To be done later.

A.2 Stacks and Strengthening

To be done later.

Draft of April 2025, comments welcome.

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